

PILLOWCASE FLOER HOMOLOGY OF THE TORUS KNOTS $T(3, n)$

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ABSTRACT. Hedden, Herald and Kirk defined a Lagrangian Floer homology in the pillowcase from a tangle decomposition of a knot and conjectured that, for a suitable decomposition, it recovers reduced singular instanton homology. We compute this homology for every torus knot $T(3, n)$, using their decomposition. For all sufficiently small holonomy perturbations in an explicit closed cone, the immersed curves are restricted and the complex has $1 + |\sigma|$ generators, where σ is the signature. There is no differential when $n \equiv 1, 2 \pmod{6}$ and exactly one, to the generator of the abelian representation, when $n \equiv 4, 5 \pmod{6}$. The $\mathbb{Z}/4$ -graded homology has the graded dimensions of reduced instanton homology. This verifies the conjecture of Hedden, Herald and Kirk for this infinite family, and, combined with a cobordism inequality of Daemi and Scaduto, shows that over $\mathbb{F}_2$ the pillowcase complex is S-chain homotopy equivalent to their singular instanton S-complex. The proof rests on an explicit description of the traceless character variety and the strict monotonicity of an angle function along its components; three inequalities are verified by interval arithmetic. For a different decomposition of $T(3, 4)$ in the same family, a numerical computation gives homology of rank 7 rather than 5 with zero bounding cochain.

1. INTRODUCTION

Hedden, Herald and Kirk proposed a Lagrangian Floer theory in the pillowcase as a model for the reduced singular instanton homology $I^\natural(K)$ of a knot $K \subset S^3$ [HHK14, HHK18]. A Conway sphere splits (S^3, K) into a trivial tangle (D, U) and a 2-tangle (Y, T) . Restriction to the four-punctured sphere sends the traceless $SU(2)$ character varieties of the two sides to immersed curves L_0 and L_1 in the pillowcase $P = R(S^2, \{a, b, c, d\})$. After a small holonomy perturbation π of (Y, T) , and with an additional meridional “earring” on the trivial side, these are restricted immersed curves in the sense of [HHK18], and their Floer homology $H^\natural(Y, T, \pi)$ is a $\mathbb{Z}/4$ -graded vector space over $\mathbb{F}_2$. Its generators correspond to the critical points of a perturbed Chern–Simons functional, and Hedden, Herald and Kirk conjecture that a suitable decomposition recovers $I^\natural(K)$ ([HHK18, Conj. 6.5], restated as Conjecture 2.5 below).

For 2-bridge knots the curve L_1 is a linear arc, all differentials vanish, and the conjecture holds [HHK18, §7]. For torus knots Hedden, Herald and Kirk computed the complex for nine knots, among them $T(3, 4)$, $T(3, 5)$ and $T(3, 7)$, using a tangle decomposition adapted to the torus knot [HHK18, §§10–11]. In these examples the homology has the rank of $I^\natural(K)$, and nonzero differentials occur. The choice of decomposition matters: with the earring placed on the other tangle the homology of $T(4, 5)$ has the wrong rank [CHKK22, §10.3].

In this paper we compute the pillowcase complex of every torus knot $T(3, n)$ in the decomposition of [HHK18, §10]. Fix $n \geq 4$ coprime to 3, write $n \equiv \eta \pmod{3}$ with $\eta = \pm 1$, and let $(r, s) = ((2n+1)/3, -2)$ if $\eta = 1$ and $(r, s) = ((n+1)/3, -1)$ if $\eta = -1$. These are the parameters used in [HHK18] for $n = 4, 5, 7$ and in [FKP17] for $n = 10$. The perturbation has two real parameters (ϵ_A, ϵ_B) , and the earring curve L_0 depends on a parameter $\epsilon > 0$ (Section 2); from Section 3 on we abbreviate (ϵ_A, ϵ_B) to ϵ . For $n \equiv 4, 5 \pmod{6}$ the unperturbed variety has two

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double points, and we require the perturbation to resolve both of them at a definite rate: (ϵ_A, ϵ_B) lies in a closed cone $\mathcal{C}_a \subset \mathbb{R}^2$, the complement of thin wedges around at most two lines through the origin (Definition 5.5). For $n \equiv 4 \pmod{6}$ the cone excludes the axis $\epsilon_B = 0$, and for $n \equiv 1, 2 \pmod{6}$ it is all of $\mathbb{R}^2$.

Theorem 1.1. *Let $K = T(3, n)$ with $n \geq 4$ coprime to 3, and let (Y, T) be the tangle above. Fix $a > 0$ small. For every sufficiently small $(\epsilon_A, \epsilon_B) \in \mathcal{C}_a \setminus \{0\}$, and every sufficiently small earring parameter $\epsilon > 0$ (chosen after the perturbation):*

- (1) L_1 is a restricted immersed 1-manifold, (L_0, L_1) is an admissible pair, and the complex $C(L_0, L_1)$ has $1 + |\sigma(K)|$ generators;
- (2) if $n \equiv 1, 2 \pmod{6}$ the differential vanishes; if $n \equiv 4, 5 \pmod{6}$ there is exactly one bigon, and it runs from a generator x^- to the generator r_+ of [HHK18, Lemma 5.1];
- (3) $H^\natural(Y, T, \pi)$ has rank $\|\Delta_K\|_1$, the sum of the absolute values of the coefficients of the Alexander polynomial. With the absolute grading normalized by $\text{gr}(r_+) = \sigma(K)$, its $\mathbb{Z}/4$ -graded dimensions are those of $I^\natural(K; \mathbb{Q})$ in Kronheimer and Mrowka's grading, whatever absolute gradings are chosen on the closed components of L_1 .

Corollary 1.2. *For every torus knot $T(3, n)$ with $n \geq 4$ coprime to 3, the decomposition above satisfies the conclusion of Conjecture 6.5 of [HHK18]: $H^\natural(Y, T, \pi)$ and $I^\natural(K; \mathbb{Q})$ have the same $\mathbb{Z}/4$ -graded dimensions.*

For $n = 4, 5, 7$ this recovers the computations of [HHK18, §11]. We also describe the curves explicitly. The arc component of L_1 lifts to a curve in the strip $0 < \gamma < \pi$ of the plane from $(0, 0)$ to $(\pi, 2\pi j)$. For $n \equiv 1, 2 \pmod{6}$ one has $j = 1, 2$, and the arc carries 1, resp. 3, generators. For $n \equiv 4, 5 \pmod{6}$ the perturbation resolves the two double points in one of two ways. In the *split* resolution $j = 1$, resp. $j = 2$, the arc carries 3, resp. 5, generators, and L_1 has one further closed component; in the *joined* resolution $j = 3$, resp. $j = 4$, and the arc carries 7, resp. 9, generators. Every closed component carries four generators, one in each degree, and no differential, and the two resolutions give the same homology (Theorem 6.10). For $n \equiv 5 \pmod{6}$ our computations show both resolutions for perturbations in $\mathcal{C}_a$, the split one in the sector containing the ϵ_B -axis and the joined one in the sector containing the ϵ_A -axis. For $n \equiv 4 \pmod{6}$ the resolution is the same for all small perturbations in $\mathcal{C}_a$, and it was split in all our computations (Remark 5.8).

The second result concerns the differential. Daemi and Scaduto associate to a knot an S -complex $\tilde{C}(K)$, a chain complex with an involution-like map χ , whose homology is $I^\natural(K)$ and whose differential has a component δ_1 to the reducible generator [DS24b]. In [Wue26c] we proposed that the pillowcase complex, with $\chi(x_i^-) = x_i^+$, is a model for $\tilde{C}(K; \mathbb{F}_2)$ (Conjecture 2.6). For torus knots the differential of $\tilde{C}(K; \mathbb{Q})$ has rank $R = \frac{1}{2}(1 + |\sigma| - \|\Delta_K\|_1)$, and for $T(3, n)$ one has $R = 1$ if $n \equiv 4, 5 \pmod{6}$ and $R = 0$ otherwise [Wue26c].

Theorem 1.3. *Let $K = T(3, n)$ with $n \geq 4$ coprime to 3, and let a , the perturbation and the earring parameter be as in Theorem 1.1. Then the conclusion of part (a) of Conjecture 2.6 holds for K and these perturbations. Normalize the absolute grading on each closed component of L_1 so that its generators x^- have odd degree after subtracting $\sigma(K)$ (Definition 6.13). Then the S -complex $\mathcal{P}(K)$, shifted by $\sigma(K)$, has the graded generators of $\tilde{C}(K)$, and its differential is a single component of type δ_1 , of rank R . Consequently, $\mathcal{P}(K)$ shifted by $\sigma(K)$ is S -chain homotopy equivalent to $\tilde{C}(K; \mathbb{F}_2)$; that is, part (b) of Conjecture 2.6 holds for K .*

The normalization in Theorem 1.3 agrees with the convention of [HHK18, §6.1] for closed components, given the identification of the relative grading of [HHK18] with Kronheimer and Mrowka's, which is outlined in [HHK18, §6.1]. For $n \equiv 1, 2 \pmod{6}$, $\tilde{C}(K; \mathbb{Z})$ has zero differential because $R = 0$. For $n \equiv 4, 5 \pmod{6}$ the input from the instanton side is that δ_1 takes an odd

value; it follows from a cobordism inequality of Daemi and Scaduto for their invariant h with coefficients in $\mathbb{F}_2$, applied to full twists on three strands (Section 6). For $T(3, 4)$ and $T(3, 5)$ they show directly that $\delta_1 = \pm 1$ on each generator of degree 1 [DS24b, §§9.3–9.4].

The decomposition appears to matter even within the family of [HHK18, §10]. Any (r, s) with $3r + ns = 1$ defines a tangle decomposition of $T(3, n)$, and different choices give different curves L_1 [HHK18, §11]. In Section 7 we report a numerical computation for $T(3, 4)$ with $(r, s) = (7, -5)$. There the pair (L_0, L_1) again appears to be admissible with L_1 restricted. The arc, however, is not homotopic relative to its ends to the arc of the decomposition above. With zero bounding cochain the complex has no bigon, and $H^\natural$ has rank 7, whereas $I^\natural(T(3, 4))$ has rank 5. The arc has a self-intersection near the corner $(0, 0)$ that defines a bounding cochain, and the deformed complex has homology of rank 5, by the mechanism found in [CHKK22, §10.3] for $T(4, 5)$. Both results are consistent with Conjecture 6.5 of [HHK18], which asserts the existence of a suitable decomposition. A similar dependence occurs for Smith's decomposition of the pretzel knot $P(-2, 3, 7)$, where the pairing of explicit model curves with zero bounding cochain has dimension 7 rather than 9 [Wue26b, Prop. 6.7]. In the other direction, regluing the tangle of [HHK18, §10] for $T(3, 5)$ by Dehn twists along the Conway sphere gives the pretzel knots $P(-2, 3, q)$, and shearing the curves of this paper proves Conjecture 6.5 of [HHK18] for all of them, and for a family of twisted torus knots [Wue26b, §§4–5].

Outline of the proof. The proof of Theorem 1.1 has four steps.

The unperturbed variety (Section 3). In HHK's coordinates (u, v, τ) the equations of [HHK18, §10] for $T(3, n)$ reduce to the pair (E1), (E2) of trigonometric equations. Eliminating τ gives $4 \cos^2 u = 1 - \sin(nv)/\sin(kv)$ with $k = (n + 2\eta)/3$. The variety consists of the arc B of binary dihedral representations and $\lfloor n/3 \rfloor$ circles over explicit intervals of the v -axis. For $n \equiv 1, 2 \pmod{6}$ these are disjoint. For $n \equiv 4, 5 \pmod{6}$ the circle over the interval containing $v = \pi/2$ meets B in two nondegenerate double points.

Monotonicity (Section 4). Let $\varphi = \theta - \gamma$ in the pillowcase coordinates (γ, θ) , so that the diagonal arc Δ , near which L_0 lies, is $\{\varphi \in 2\pi\mathbb{Z}\}$ in the plane. We show that φ is strictly monotone along every circle that avoids $v = \pi/2$, and along each half of the central circle. In the variables $\psi = mv$, $\delta = \eta v$, an exact trigonometric identity expresses $(d\varphi/d\psi)^2$ through a function E that is linear in $1/m$. Its nonvanishing then reduces to the positivity of an explicit function of two variables on a compact domain independent of n . We verify that positivity by a rigorous interval-arithmetic computation (Appendix A). The same analysis gives the total change of φ along each piece and shows that the central circle stays away from the edges of the pillowcase.

Small perturbations (Section 5). The perturbation is controlled in three regions. Near the two abelian representations a chart in which the abelian locus is $\{w = 0\}$ shows that the perturbed arc ends at the corners along a half-curve, uniformly in ϵ , with limiting slope close to that of B . Near the double points a Morse model shows that they are resolved, and that the image of each resolution is immersed. Elsewhere the perturbed variety is C^1 -close to the unperturbed one.

The complex (Section 6). Generators are the transverse crossings of the lifted curves with the lines $\Delta_k = \{\varphi = 2\pi k\}$, together with r_+ . A bigon lifts to the plane, and its boundary on L_1 must contain a critical point of φ ; otherwise its two corners would lie over a single crossing on different lifts of L_0 . Along the arc, φ has two critical points, at the resolved double points, and the values of φ there show that only one pair of generators can bound a bigon. For $n \equiv 4, 5 \pmod{6}$ that pair does bound an embedded bigon near the corner $(0, 0)$. On stretches where φ is monotone the Maslov term of the grading formula vanishes, which gives the gradings. Comparing with the graded ranks of $I^\natural(K)$ computed in [Wue26c] proves part (3).

Everything except the three inequalities of Appendix A is proved by hand. For many values of n we have also computed the perturbed complexes numerically, and they agree with the theorems (Remark 6.12).

Organization. Section 2 reviews the pillowcase Floer theory of [HHK18] and the S-complexes of Daemi and Scaduto. Section 3 describes the unperturbed variety, Section 4 proves the monotonicity statements, and Section 5 treats small perturbations. Section 6 computes the complex and proves Theorems 1.1 and 1.3, and Section 7 treats the second decomposition of $T(3, 4)$. Appendix A contains the interval-arithmetic verifications.

Conventions. Positive torus knots have negative signature, as in [HHK18]; thus $\sigma(T(3, 4)) = -6$. We cite results of [HHK18] by the numbering of arXiv:1501.00028v1.

2. PILLOWCASE FLOER HOMOLOGY

This section collects the parts of Hedden, Herald and Kirk's Lagrangian Floer theory in the pillowcase [HHK18] that we use, together with Daemi and Scaduto's S-complexes [DS24b]. Results of [HHK18] are cited by their numbering in arXiv:1501.00028v1. Coefficients in Floer complexes are in $\mathbb{F}_2$ throughout.

2.1. The pillowcase. Let $G \cong \mathbb{Z}^2 \rtimes \mathbb{Z}/2$ be the group of isometries of $\mathbb{R}^2$ generated by $(\gamma, \theta) \mapsto (\gamma + 2\pi, \theta)$, $(\gamma, \theta) \mapsto (\gamma, \theta + 2\pi)$ and $(\gamma, \theta) \mapsto (-\gamma, -\theta)$. The *pillowcase* is the quotient $P = \mathbb{R}^2/G$, a topological 2-sphere; the quotient map $\mathbb{R}^2 \rightarrow P$ is a branched covering, branched over the images of the lattice points $(\pi\mathbb{Z})^2$, which are the four *corners* of P [HHK18, §3.1]. The rectangle $[0, \pi] \times [0, 2\pi]$ is a fundamental domain; we call $[0, \pi] \times [0, \pi]$ the front face, $[0, \pi] \times [\pi, 2\pi]$ the back face, and the images of the lines $\gamma \in \pi\mathbb{Z}$ the left and right edges. Put $P^* = (\mathbb{R}^2 \setminus (\pi\mathbb{Z})^2)/G$, oriented by $d\gamma \wedge d\theta$. The restriction

$$\mathbb{R}^2 \setminus (\pi\mathbb{Z})^2 \longrightarrow P^*$$

is a regular covering with deck group G . Paths, loops and immersed disks in P^* therefore lift to $\mathbb{R}^2 \setminus (\pi\mathbb{Z})^2$, and we shall work with such lifts throughout.

The pillowcase is the traceless character variety of the four-punctured sphere: the map $\psi(\gamma, \theta): a \mapsto \mathbf{i}, b \mapsto e^{\gamma\mathbf{k}}\mathbf{i}, c \mapsto e^{\theta\mathbf{k}}\mathbf{i}, d \mapsto e^{(\theta-\gamma)\mathbf{k}}\mathbf{i}$ induces a homeomorphism of P with $R(S^2, \{a, b, c, d\})$, the space of conjugacy classes of $\mathrm{SU}(2)$ representations of $\pi_1(S^2 \setminus \{a, b, c, d\}) = \langle a, b, c, d \mid ba = cd \rangle$ sending each of a, b, c, d to a traceless element, and the corners correspond to the abelian representations [HHK18, Prop. 6.1], [HHK14, Prop. 3.1].

On $\mathbb{R}^2$ we use the function

$$\varphi = \theta - \gamma.$$

Translations in G change φ by elements of $2\pi\mathbb{Z}$, and $(\gamma, \theta) \mapsto (-\gamma, -\theta)$ changes its sign. The *diagonal arc* $\Delta = \{(\gamma, \gamma) \mid \gamma \in [0, \pi]\} \subset P$ [HHK18, (12)] joins the corners $(0, 0)$ and (π, π) , and its preimage in $\mathbb{R}^2$ is the union of the lines

$$\Delta_k = \{\varphi = 2\pi k\}, \quad k \in \mathbb{Z}.$$

The lines $\{\varphi = \pi + 2\pi k\}$ are lifts of the arc of slope one on the back face, which is not part of Δ . Every lattice point lies on a line $\{\varphi \in \pi\mathbb{Z}\}$; the lattice points on Δ_k are the points $(j\pi, j\pi + 2\pi k)$.

2.2. The earring curve. For $\epsilon > 0$ let $L_0 = L_0^{\epsilon, 0}: S^1 \rightarrow P^*$ be the immersion defined by

$$\tilde{L}_0(t) = \left(t + \epsilon \sin t + \frac{\pi}{2}, t - \epsilon \sin t + \frac{\pi}{2}\right), \quad t \in [0, 2\pi], \quad (2.1)$$

followed by the branched covering [HHK18, Def. 3.4, (13)]; as in [HHK18, §11] we take the perturbation function g of [HHK18, (13)] to be zero. It is the restriction to the boundary of the traceless character variety of the trivial tangle with an earring, after a small holonomy perturbation [HHK18, Thm. 6.2]. For small ϵ it is self-transverse with one double point, at $(\pi/2, \pi/2)$; it lies in an $O(\epsilon)$ -neighbourhood of Δ and passes around the two corners $(0, 0)$ and (π, π) through the back face. It is an unobstructed circle with $\mu(L_0, \ell_1) = 0 = z(L_0)$ in the notation of §2.3 [HHK18, pp. 16, 18].

Since $\tilde{L}_0(t + 2\pi) = \tilde{L}_0(t) + (2\pi, 2\pi)$, the preimage of L_0 in $\mathbb{R}^2$ is the union of the curves

$$S_+^k = \tilde{L}_0 + (0, 2\pi k), \quad S_-^k = -\tilde{L}_0 + (0, 2\pi k), \quad k \in \mathbb{Z},$$

which we call the *strands* of L_0 ; each is a properly embedded line lying within $\sqrt{2}\epsilon$ of Δ_k , and a translation by $(2\pi a, 2\pi b)$ carries $S_\pm^k$ to $S_\pm^{k+b-a}$. Reparametrizing, $S_-^0 = \{(t + \frac{\pi}{2} - \epsilon \sin t, t + \frac{\pi}{2} + \epsilon \sin t)\}$. The slope of S_+ at the point with parameter t is $(1 - \epsilon \cos t)/(1 + \epsilon \cos t)$, and that of S_- is its reciprocal. Hence on the front strip $0 < \gamma < \pi$ the strand S_+^k has slope less than 1, and S_-^k has slope greater than 1 for $\epsilon < \gamma < \pi - \epsilon$ (it has slope 1 at $\gamma = \epsilon$ and $\gamma = \pi - \epsilon$); in P these are the two branches of L_0 along the interior of Δ [HHK18, §5.1]. The two strands near Δ_k cross each other exactly at the points $(\frac{\pi}{2} + j\pi, \frac{\pi}{2} + j\pi + 2\pi k)$, and they pass each lattice point of Δ_k on opposite sides at distance $\sqrt{2}\epsilon$: for instance S_+^0 passes $(0, 0)$ through $(-\epsilon, \epsilon)$ and S_-^0 through $(\epsilon, -\epsilon)$, and the sides alternate from one lattice point of Δ_k to the next.

2.3. Restricted curves and admissible pairs. Let ℓ_1 be the line field of slope one on $\mathbb{R}^2$; it descends to P^* . For a path α immersed in P^* , the Maslov index $\mu(\alpha, \ell_1)$ is the signed number of times the tangent line of α passes ℓ_1 , counterclockwise passages counting +1, with the half-open endpoint convention of [HHK18, §2.3]: if $\vartheta(t)$ is a continuous choice of the angle of the tangent line minus $\pi/4$, then $\mu = \lfloor (\vartheta(1) - \varsigma)/\pi \rfloor - \lfloor (\vartheta(0) - \varsigma)/\pi \rfloor$ for small $\varsigma > 0$. For two lines ℓ_0, ℓ'_1 and a line ℓ , all pairwise transverse, the *triple index* $\tau(\ell_0, \ell'_1, \ell) \in \{0, 1\}$ is 1 exactly when the clockwise rotation of ℓ_0 to ℓ'_1 passes through ℓ [HHK18, Def. 2.5]. Let $z \in H^1(P^*; \mathbb{Z}/4)$ be the class assigning 1 to a small loop around a corner, oriented counterclockwise [HHK18, Def. 3.1]. A loop winding once around the pillowcase in the open strip between the two edges encircles two corners, so z takes the value 2 on it.

An immersed circle in P^* is *unobstructed* if every lift of it to the universal cover of P^* is a properly embedded line; by a lemma of Abouzaid [Abo08, Lemma 2.2] this holds if and only if it is homotopically essential and contains no *fishtail*, that is, no subinterval whose endpoints map to the same point and which maps to a nullhomotopic loop [HHK18, Def. 2.1]. A *proper immersion* of an interval is the image of a smooth immersion $I \rightarrow \mathbb{R}^2$ sending the endpoints to lattice points and the interior into $\mathbb{R}^2 \setminus (\pi\mathbb{Z})^2$; its slopes at the endpoints are its *limiting slopes* [HHK18, Def. 3.3].

Definition 2.1 ([HHK18, Def. 3.6 and p. 33]). A *restricted immersed circle* is an unobstructed immersed circle $L: S^1 \rightarrow P^*$ with $\mu(L, \ell_1) + z(L) \equiv 0 \pmod{4}$. A *restricted immersed arc* is a proper immersion of an interval whose restriction to the complement of small discs around the corners contains no fishtail. A *restricted immersed 1-manifold* is a disjoint union of one restricted immersed arc and finitely many restricted immersed circles.

The image of a straight segment joining two lattice points whose interior misses the lattice is a restricted immersed arc [HHK18, p. 18].

Definition 2.2 ([HHK18, Def. 3.11]). A pair (L_0, L_1) of restricted immersed curves is *admissible* if at least one of them is a circle, if L_0 and L_1 intersect transversely, and if, whenever loops α_0 in the domain of L_0 and α_1 in the domain of L_1 have $L_0 \circ \alpha_0$ freely homotopic to $L_1 \circ \alpha_1$ in P^* , both α_0 and α_1 are nullhomotopic.

2.4. The complex and its grading. For an admissible pair, $C(L_0, L_1)$ is the $\mathbb{F}_2$ -vector space on $L_0 \cap L_1$. A *bigon* from p to q is an orientation-preserving immersion of a disc into P^* with two convex corners, at p and q , whose boundary, traversed counterclockwise, runs along L_0 from p to q and then along L_1 from q back to p ; equivalent bigons are identified [HHK18, Def. 2.10, (5)]. The differential is $\partial p = \sum_q \# \mathcal{M}(p, q) q$, the count of bigons from p to q modulo 2. By [HHK18, Cor. 3.13] each $\mathcal{M}(p, q)$ contains at most one bigon. A bigon lifts to an immersed disc in $\mathbb{R}^2 \setminus (\pi\mathbb{Z})^2$, so its boundary lies on one component of L_1 and the two generators lie on the same component; in particular there is no differential between generators on different components of L_1 [HHK18, p. 37]. We write $HF(L_0, L_1)$ for the homology.

The complex carries a relative $\mathbb{Z}/4$ -grading $\text{gr}(p, q) = \text{gr}(p) - \text{gr}(q)$, defined for generators on the same component of L_1 [HHK18, Def. 3.7]; the differential lowers it by one, and a bigon from p to q forces $\text{gr}(p, q) = 1$ [HHK18, Prop. 3.8]. It is computed as follows [HHK18, Lemma 3.10]: if α_0 is a path in L_0 from p to q and α_1 a path in L_1 from q to p , then

$$\text{gr}(p, q) = \mu(\alpha_0, \ell_1) + \mu(\alpha_1, \ell_1) + \tau(T_p L_0, T_p L_1, \ell_1) - \tau(T_q L_0, T_q L_1, \ell_1) + z(\alpha_0 * \alpha_1) \pmod{4}. \quad (2.2)$$

Two consequences are recorded in [HHK18, §§5.1–5.2].

- (1) If L_1 meets the interior of Δ transversely at x , then for small ϵ there are exactly two generators near x : x^+ on the branch of L_0 of slope less than 1 and x^- on the branch of slope greater than 1. They satisfy $\text{gr}(x^+, x^-) = 1$.
- (2) If p_+, q_+ lie on the branch of slope less than 1 on the front face, α_0 runs along that branch, and L_1 is transverse to Δ at both points, then

$$\text{gr}(p_+, q_+) = \mu(\alpha_1, \ell_1) + z(\alpha_0 * \alpha_1) \pmod{4}, \quad (2.3)$$

and the same formula holds for two points on the branch of slope greater than 1.

In terms of lifts, $x^\pm$ are the intersections of a lift of L_1 with the strands $S_\pm^k$ near a point of Δ_k on the front strip.

Suppose L_1 is a restricted immersed arc with an endpoint at the corner $(0, 0)$ whose limiting slope is bounded away from 1. Then for small ϵ there is a unique intersection point r_+ of L_0 and L_1 converging to that corner as $\epsilon \rightarrow 0$ [HHK18, Lemma 5.1]. When L_1 comes from a knot K as in §2.5, the grading of the component containing the arc is made absolute by

$$\text{gr}(r_+) = \sigma(K) \pmod{4}, \quad (2.4)$$

[HHK18, (19)], with the convention that positive torus knots have negative signature. On circle components HHK fix the absolute grading by declaring one generator to have the grading assigned to the corresponding generator of Kronheimer and Mrowka's complex [HHK18, §6.1], [KM11b, Prop. 4.4].

Theorem 2.3 ([HHK18, Thm. 5.4]). *Let $L_1: S^1 \rightarrow P^*$ be a restricted immersed circle that misses the left and right edges of the pillowcase, and let $\tilde{L}_1: [0, 2\pi] \rightarrow \mathbb{R}^2$ be a lift of $t \mapsto L_1(e^{it})$. Then its vertical degree $d = \frac{1}{2\pi} |\theta(\tilde{L}_1(2\pi)) - \theta(\tilde{L}_1(0))|$ is even, and $HF(L_0, L_1)$ has rank $d/2$ in each of the four gradings.*

Remark 2.4. By [HHK18, Thm. 4.1, Cor. 4.6], $HF(L_0, L_1)$ is independent of ϵ and, as a relatively graded group, depends only on the free homotopy classes of the circle components of L_1 and the homotopy class rel endpoints of its arc; Theorem 2.3 is deduced from [HHK18, Cor. 4.6] in [HHK18, p. 33]. The proofs of our main results use neither: the computations in Section 6 are

direct. The invariance serves as a consistency check in Section 6 and in the numerical Section 7, and Theorem 2.3 is quoted only in Corollary 4.8, which is not used later.

2.5. Tangle decompositions. Let $K \subset S^3$ be a knot, decomposed along a Conway sphere as $(S^3, K) = (Y, T) \cup (D, U)$, where (D, U) is the trivial 2-tangle in a ball and the boundary is identified with $(S^2, \{a, b, c, d\})$ [HHK18, (17)]. Let $R(Y, T)$ be the space of conjugacy classes of traceless $SU(2)$ representations of $\pi_1(Y \setminus T)$ and $R_\pi(Y, T)$ its version perturbed by a holonomy perturbation π supported in the interior of Y . Restriction to the boundary gives $L_1 : R_\pi(Y, T) \rightarrow P$. The earring curve L_0 of §2.2 comes from the trivial side. When L_1 is a restricted immersed 1-manifold and (L_0, L_1) is admissible,

$$H^\natural(Y, T, \pi) = HF(L_0, L_1),$$

which by [HHK18, p. 34, §6.1] is the direct sum of the contributions of the components of $R_\pi(Y, T)$. The two abelian traceless representations of $\pi_1(Y \setminus T)$ are the endpoints of the arc component; the one that extends over the knot complement maps to the corner $(0, 0)$, and its perturbation gives the generator r_+ [HHK18, §6.1, §8]. The remaining generators come in pairs $x^\pm$, one pair for each transverse intersection point of L_1 with the interior of Δ . For a torus knot, with the tangle side unperturbed and a small perturbation on the side of the earring, these intersection points correspond to the irreducible traceless representations of the knot group, and there are $1 + |\sigma(K)|$ generators [HHK14, Thm. 11.1], [HHK18, §11.10]. For $T(3, n)$, Theorem 1.1 shows that this count persists after a small generic perturbation of the tangle side. Hedden, Herald and Kirk conjecture:

Conjecture 2.5 ([HHK18, Conj. 6.5]). “Given a knot (X, K) in a homology 3-sphere, there exists a 2-tangle decomposition as in Equation (17), such that for suitably small generic perturbations π , $L_1 : R_\pi(Y, T) \rightarrow P$ is a restricted immersed 1-manifold and $H^\natural(Y, T, \pi)$ is isomorphic to the reduced instanton homology $I^\natural(X, K)$.”

The conjecture asserts the existence of one good decomposition. The answer does depend on the decomposition: different choices of the integers r, s of §3 give different curves L_1 [HHK18, p. 55], and with the earring on the other tangle the pillowcase homology of $T(4, 5)$ has the wrong rank [CHKK22, §10.3].

2.6. S-complexes. Daemi and Scaduto refine the chain complex of reduced singular instanton homology to an S -complex [DS24b, Def. 3.21, §4.1]

$$\tilde{C}_*(K) = C_*(K) \oplus C_{*-1}(K) \oplus \mathbb{Z}, \quad \tilde{d} = \begin{pmatrix} d & 0 & 0 \\ v & -d & \delta_2 \\ \delta_1 & 0 & 0 \end{pmatrix},$$

where $C_*(K)$ is generated by irreducible critical points of the singular Chern–Simons functional, the summand $\mathbb{Z}$ in degree 0 by the reducible one, δ_1 and δ_2 count instantons to and from the reducible, and $v : C_* \rightarrow C_{*-2}$ is defined by holonomy maps on one-dimensional moduli spaces; degrees are read modulo 4. The S -structure is $\chi(\alpha, \beta, n) = (0, \alpha, 0)$, which satisfies $\chi^2 = 0$ and $\chi\tilde{d} + \tilde{d}\chi = 0$. Morphisms of S -complexes are degree-0 chain maps commuting with χ and inducing the identity on the reducible summand, and S -chain homotopy equivalences are defined accordingly [DS24b, Defs. 3.28, 3.30]. The S -chain homotopy type of $\tilde{C}(K)$ is an invariant of K , its homology is $I^\natural(K)$ up to a shift of grading [DS24b, Thm. 3.34, Thm. 8.9], and its local equivalence class carries the integer invariant $h(K)$; for an S -complex with $d = 0$, $\delta_1 \neq 0$ if and only if $h > 0$ [DS24b, Prop. 4.15, Rem. 4.2].

For a torus knot K the traceless character variety is finite and nondegenerate, with $|\sigma|/2$ irreducible points and one abelian point, and the S -complex can be built on it: then $d = 0$, $\delta_2 = 0$

and $C_*(K) = C_1 \oplus C_3$ with

$$\text{rank } C_1 = N_1 = \lceil |\sigma|/4 \rceil, \quad \text{rank } C_3 = N_3 = \lfloor |\sigma|/4 \rfloor$$

[DS24a, §1, Cor. 3], [DS24b, §9.5], [Wue26c]; for $T(3, n)$ the generators, gradings and differential of $\tilde{C}(K)$ are assembled in [Wue26a]. In $\tilde{C}$ the group C_1 occurs in degrees 1 and 2 and C_3 in degrees 3 and 0. Since $I^\natural(K)$ has rank $\|\Delta_K\|_1$, the sum of the absolute values of the coefficients of the Alexander polynomial [KM11b, Prop. 1.4], [LY22, Cor. 1.8], the differential $\tilde{d}$ has rank

$$R = \frac{1}{2}(1 + |\sigma(K)| - \|\Delta_K\|_1)$$

over $\mathbb{Q}$, and in the degrees of $\tilde{C}(K; \mathbb{Q})$, with the reducible in degree 0, the group $I^\natural(K; \mathbb{Q})$ has rank vector

$$(N_3 + 1 - \lceil R/2 \rceil, N_1 - \lceil R/2 \rceil, N_1 - \lfloor R/2 \rfloor, N_3 - \lfloor R/2 \rfloor) \quad (2.5)$$

in degrees 0, 1, 2, 3 [Wue26c]. Kronheimer and Mrowka's absolute grading is obtained by adding $\sigma(K)$ modulo 4 [Wue26c], [DS24b, Thm. 8.9].

For small perturbations the pillowcase complex of a torus knot has $1 + |\sigma|$ generators: r_+ , in grading $\sigma \bmod 4$, and a pair x_i^-, x_i^+ with $\text{gr } x_i^+ = \text{gr } x_i^- + 1$ for each irreducible traceless character, the two points over the corresponding intersection with Δ . If every bigon runs from some x_i^- either to r_+ or to some x_j^+ with $j \neq i$, then

$$\chi(x_i^-) = x_i^+, \quad \chi(x_i^+) = \chi(r_+) = 0$$

make the pillowcase complex an S-complex $\mathcal{P}(K)$ over $\mathbb{F}_2$ with reducible generator r_+ , in which x_i^- and x_i^+ play the roles of the two copies of an irreducible generator [Wue26c].

Conjecture 2.6 ([Wue26c]). *Let $K = T(p, q)$, and consider the pillowcase complex of [HHK18, §§10–11] over $\mathbb{F}_2$, for the decomposition in which the earring lies on the trivial tangle and the integers r, s with $pr + qs = 1$ satisfy $0 < r < q$, with zero bounding cochains and a holonomy perturbation small enough that the complex has exactly the $1 + |\sigma|$ generators $r_+, x_i^\pm$.*

- (a) *Every bigon runs from some x_i^- either to r_+ or to some x_j^+ with $j \neq i$.*
- (b) *The S-complex $\mathcal{P}(K)$, with σ subtracted from its grading, is S-chain homotopy equivalent to $\tilde{C}(K; \mathbb{F}_2)$.*

The decompositions of [HHK18, §§10–11] depend on integers (r, s) with $pr + qs = 1$. The normalization $0 < r < q$ determines them, since $pr \equiv 1 \pmod{q}$, and it is satisfied by all the choices printed in [HHK18, §11] and by those of Theorem 1.1. Section 7 gives numerical evidence that it is needed: for $T(3, 4)$ with $(r, s) = (7, -5)$ and zero bounding cochains, the homology has rank 7 rather than 5.

Since h is an invariant of the S-chain homotopy type, part (b) predicts that the component of the pillowcase differential with values in the span of r_+ is nonzero if and only if $\delta_1 \otimes \mathbb{F}_2 \neq 0$ [Wue26c].

3. THE TRACELESS CHARACTER VARIETY OF THE TANGLE

This section describes the unperturbed variety $W = W_0$ for $K = T(3, n)$ completely: its components, the image of the binary dihedral arc in the pillowcase, and the two double points that occur when $n \equiv 4, 5 \pmod{6}$. For $n = 4, 5, 7$ and 10 the description is in [HHK14, §11], [HHK18, §11] and [FKP17, §4]. Recall that $n = 3m + \eta$ with $\eta \in \{\pm 1\}$, and that $m' = 2m + \eta$, $k = m + \eta$, $N = n + k = 2m'$ and $\nu = \lfloor n/3 \rfloor$; thus $\nu = m$ if $\eta = 1$, $\nu = m - 1$ if $\eta = -1$, and $N = 4\nu + 2$ in both cases.

3.1. The tangle and its equations. Let p, q be coprime and fix integers r, s with $pr + qs = 1$. Hedden, Herald and Kirk decompose $(S^3, T(p, q))$ along a Conway sphere into a trivial tangle (D, U) and a tangle (Y, T) in a 3-ball [HHK18, §10]. The group $\pi_1(Y \setminus T)$ is free on two elements A_1, B_1 [HHK14, §11], and the holonomy perturbations are supported near two curves P_A, P_B in Y , with perturbation functions $\epsilon_A \sin$ and $\epsilon_B \sin$. A perturbed traceless representation is determined, up to conjugation, by

$$A_1 \mapsto e^{uQ_A}, \quad B_1 \mapsto e^{vQ_B}, \quad Q_A = \mathbf{i}, \quad Q_B = \tau \mathbf{i} + \sqrt{1 - \tau^2} \mathbf{j},$$

with $(u, v, \tau) \in [0, \pi]^2 \times [-1, 1]$. The condition that the meridians a, b of T be sent to traceless elements is $\Psi = (\Psi_1, \Psi_2) = 0$, where [HHK18, (33)–(34)]

$$\Psi_i = \cos \Theta_i \cos \Phi_i - \tau \sin \Theta_i \sin \Phi_i \quad (i = 1, 2), \quad (3.1)$$

$$\begin{aligned} \Phi_1 &= (s + p)u + (q - r)\epsilon_A \sin u, & \Theta_1 &= (q - r)v - (s + p)\epsilon_B \sin v, \\ \Phi_2 &= su - r\epsilon_A \sin u, & \Theta_2 &= -rv - s\epsilon_B \sin v. \end{aligned}$$

Write $W_\epsilon = \Psi^{-1}(0) \subset [0, \pi]^2 \times [-1, 1]$ for $\epsilon = (\epsilon_A, \epsilon_B)$, and $W = W_{(0,0)}$. By [HHK18, Thm. 10.2] there is a surjection $W_\epsilon \rightarrow R_\pi(Y, T)$. Its fibre over the image of (u, v, τ) is that single point unless $\sin u = 0$ or $\sin v = 0$, in which case it is the segment $\{(u, v)\} \times [-1, 1]$. Interior points of the box map to non-abelian representations and boundary points to abelian ones.

The restriction to the boundary sphere sends a representation to a point of the pillowcase, as follows. Conjugate so that $a \mapsto \mathbf{i}$ and $b \mapsto e^{\gamma \mathbf{k}} \mathbf{i}$ with $\gamma \in [0, \pi]$; then $c \mapsto e^{\theta \mathbf{k}} \mathbf{i}$, and (γ, θ) is the image [HHK18, Prop. 6.1], [HHK14, Thm. 11.1]. We write $\Pi: W_\epsilon \rightarrow P$ for this map. At $\epsilon = 0$, writing $\rho_A = e^{uQ_A}$ and $\rho_B = e^{vQ_B}$ for the images of A_1 and B_1 , the relevant words are [HHK14, (42)]

$$a = \rho_A^{s+p} \rho_B^{q-r}, \quad b = \rho_B^{-r} \rho_A^s, \quad c = \rho_B^{-r} a \rho_B^r. \quad (3.2)$$

From now on $p = 3$, $q = n$, and $(r, s) = (m', -2)$ if $\eta = 1$, $(r, s) = (m, -1)$ if $\eta = -1$. Then

$$(s + p, q - r, r, s) = \begin{cases} (1, m, m', -2), & \eta = 1, \\ (2, m', m, -1), & \eta = -1. \end{cases}$$

Lemma 3.1 (Normal form). *At $\epsilon = 0$, a point (u, v, τ) of the box lies in W if and only if*

$$\cos(mv) \cos u = \tau \sin(mv) \sin u, \quad (E1)$$

$$\cos(m'v) \cos 2u = \tau \sin(m'v) \sin 2u. \quad (E2)$$

More precisely, $\Psi = (E1, E2)$ if $\eta = 1$ and $\Psi = (E2, E1)$ if $\eta = -1$, where $(E1)$ and $(E2)$ also denote the differences of the two sides.

Proof. Substitute the values of $(s + p, q - r, r, s)$ into (3.1) and use that $\cos$ is even and $\sin$ is odd. □

For later use we record the perturbed phases in this notation. For $\eta = 1$ they are $\Phi_1 = u + m\epsilon_A \sin u$, $\Theta_1 = mv - \epsilon_B \sin v$, $\Phi_2 = -2u - m'\epsilon_A \sin u$ and $\Theta_2 = -m'v + 2\epsilon_B \sin v$. For $\eta = -1$ they are $\Phi_1 = 2u + m'\epsilon_A \sin u$, $\Theta_1 = m'v - 2\epsilon_B \sin v$, $\Phi_2 = -u - m\epsilon_A \sin u$ and $\Theta_2 = -mv + \epsilon_B \sin v$.

3.2. Elimination and the domain. Put

$$Q(v) = \frac{\sin(nv)}{\sin(kv)}, \quad S(v) = \frac{\sin(mv)}{\sin(kv)}, \quad x = \cos u.$$

Lemma 3.2. $1 - Q = -2 \cos(m'v) S$ and $Q - (1 - 4 \sin^2(mv)) = 2 \cos v S$.

Proof. Since $n + k = 2m'$ and $n - k = 2m$, the sum-to-product formula gives $\sin(nv) - \sin(kv) = 2 \cos(m'v) \sin(mv)$, which is the first identity. For the second, $4 \sin^2(mv) = 4S \sin(mv) \sin(kv) = 2S(\cos v - \cos(m'v))$, because $m - k = -\eta$ and $m + k = m'$; add this to the first identity. $\square$ $\square$

Lemma 3.3 (Elimination). *On the set where $\sin(mv) \sin(kv) \sin u \neq 0$, a point lies in W if and only if*

$$4x^2 = 1 - Q(v) \quad \text{and} \quad \tau = \cot(mv) \cot u.$$

For such a point, $\tau^2 = \cot^2(mv) \frac{1-Q}{3+Q}$, and $\tau^2 \leq 1$ holds if and only if $\cos v S(v) \geq 0$.

Proof. Equation (E1) is $\tau = \cot(mv) \cot u$. Substituting into (E2) and multiplying by $\sin(mv)$ gives $2x^2 \sin((m - m')v) - \cos(m'v) \sin(mv) = 0$, that is $2x^2 \sin(kv) = -\cos(m'v) \sin(mv)$, and by Lemma 3.2 this is $4x^2 = 1 - Q$. Then $\cot^2 u = x^2/(1 - x^2) = (1 - Q)/(3 + Q)$, and $\tau^2 \leq 1$ becomes $Q \geq 1 - 4 \sin^2(mv)$, which by Lemma 3.2 is $\cos v S \geq 0$. $\square$ $\square$

Accordingly let D_n be the closure in $(0, \pi)$ of the set

$$\{v \in (0, \pi) : \sin(mv) \sin(kv) \neq 0, \cos(m'v) S(v) < 0 \text{ and } \cos v S(v) > 0\}.$$

Where $\sin(mv) \sin(kv) \neq 0$, the conditions $\cos(m'v) S \leq 0$ and $\cos v S \geq 0$ are $x^2 \geq 0$ and $\tau^2 \leq 1$; together they force $Q \geq -3$, i.e. $x^2 \leq 1$.

Proposition 3.4. *The set D_n is invariant under $v \mapsto \pi - v$, and*

$$D_n \cap (0, \frac{\pi}{2}) = \bigcup_{1 \leq l \leq \lceil \nu/2 \rceil} \left[\frac{(4l-3)\pi}{N}, \frac{(4l-1)\pi}{N} \right] \cap (0, \frac{\pi}{2}).$$

Hence D_n consists of exactly ν intervals of positive length. If ν is even, these are $J_l = [(4l - 3)\pi/N, (4l - 1)\pi/N]$ and $\pi - J_l$ for $1 \leq l \leq \nu/2$, and $\pi/2 \notin D_n$. If ν is odd, they are J_l and $\pi - J_l$ for $1 \leq l \leq (\nu - 1)/2$, and the central interval $J_0 = [(2\nu - 1)\pi/N, (2\nu + 3)\pi/N]$, which contains $\pi/2 = (2\nu + 1)\pi/N$ in its interior. A point v with $\sin(mv) \sin(kv) \neq 0$ lies in D_n if and only if $\cos(m'v) S(v) \leq 0$ and $\cos v S(v) \geq 0$.

Proof. Since $m - k = -\eta$ is odd and m' is odd, $S(\pi - v) = -S(v)$, $\cos(m'(\pi - v)) = -\cos(m'v)$ and $\cos(\pi - v) = -\cos v$, so both products in the definition of D_n , and the condition $\sin(mv) \sin(kv) \neq 0$, are invariant, and D_n is symmetric. On $(0, \pi/2)$ we have $\cos v > 0$, so $D_n \cap (0, \pi/2)$ is the closure in $(0, \pi/2)$ of $\{S > 0\} \cap \{\cos(m'v) < 0\} \cap (0, \pi/2)$, and it suffices to show that $\cos(m'v) > 0$ wherever $S \leq 0$ or S is undefined, except at $v = \pi/2$. The zeros $j\pi/m$ of $\sin(mv)$ and $j\pi/k$ of $\sin(kv)$ interlace, since $|m - k| = 1$. In $(0, \pi/2)$, $S < 0$ exactly on the open intervals between $j\pi/\max(m, k)$ and $j\pi/\min(m, k)$ with $1 \leq j \leq \min(m, k)/2$, and the points where S vanishes ($j\pi/m$) or is undefined ($j\pi/k$) are the endpoints of these intervals. Since $m' = 2m + \eta = 2k - \eta$,

$$m' \cdot \frac{j\pi}{m} = 2j\pi + \eta \frac{j\pi}{m}, \quad m' \cdot \frac{j\pi}{k} = 2j\pi - \eta \frac{j\pi}{k},$$

so on the closure of such an interval $|m'v - 2j\pi| \leq j\pi/\min(m, k) \leq \pi/2$, with equality only at $v = \pi/2$. Thus $\cos(m'v) > 0$ there, except at $v = \pi/2$, and $D_n \cap (0, \pi/2)$ is the closure in $(0, \pi/2)$ of $\{\cos(m'v) < 0\} \cap (0, \pi/2)$, which is the stated union because $N = 2m'$. The interval with $4l - 1 = 2\nu + 1$ exists exactly when ν is odd, and then it ends at $\pi/2$ and merges with its mirror image into J_0 . When ν is even, the last interval in $(0, \pi/2)$ ends at $(2\nu - 1)\pi/N$, and its mirror image begins at $(2\nu + 3)\pi/N$, so $\pi/2 = (2\nu + 1)\pi/N$ is not in D_n . For the last statement, let $\sin(mv) \sin(kv) \neq 0$. Then $v \neq \pi/2$, since one of m, k is even, so $\cos v \neq 0$ and $S(v)$ is finite and nonzero; the two non-strict conditions are closed on this set, which gives one implication. Conversely, if they hold, then $\cos v S > 0$, and either $\cos(m'v) S < 0$ or v is a zero of $\cos(m'v)$, that is, an odd multiple of π/N ; every such multiple in $(0, \pi)$ other than $\pi/2$ is an endpoint of one of the intervals above. $\square$ $\square$

At an endpoint $v_f \neq \pi/2$ of one of these intervals, $\cos(m'v_f) = 0$, so $Q(v_f) = 1$, $x = 0$, $u = \pi/2$ and $\tau = 0$. Differentiating the first identity of Lemma 3.2,

$$Q'(v_f) = -2m' \sin(m'v_f) S(v_f) \neq 0, \quad (3.3)$$

since $S(v_f)$ is finite and nonzero there. We call these points the *folds*.

3.3. The components of W .

Definition 3.5. The *binary dihedral arc* is

$$B = \begin{cases} \{(\frac{\pi}{2}, \frac{\pi}{2})\} \times [-1, 1] & m \text{ even } (n \equiv 1, 5 \pmod{6}), \\ \{(u, \frac{\pi}{2}, 0) : u \in [0, \pi]\} & m \text{ odd } (n \equiv 2, 4 \pmod{6}). \end{cases}$$

For an interval J of positive length in D_n , the subset of W lying over J is

$$C_J = \left\{ (u, v, \tau) : v \in J, \cos u = \pm \frac{1}{2} \sqrt{1 - Q(v)}, \tau = \cot(mv) \cot u \right\},$$

the two signs being the two *branches*. We write $C_l = C_{J_l}$ and $C'_l = C_{\pi - J_l}$ for the *non-central circles*, and $C_0 = C_{J_0}$ for the *central circle* (when ν is odd). Its two *halves* are $C_0^L = C_0 \cap \{v \leq \pi/2\}$ and $C_0^R = C_0 \cap \{v \geq \pi/2\}$.

The representations in B are binary dihedral [FKP17, §§3–4]. For $T(3, 4)$ and $T(3, 5)$, B is the arc I_0 of [HHK18, §§11.6–11.7].

Proposition 3.6. *As a set, W is the union of B , the $2\lfloor \nu/2 \rfloor$ non-central circles, the central circle C_0 when ν is odd, and, when m is odd, the two segments $\{(u, \pi/2)\} \times [-1, 1]$ with $u \in \{0, \pi\}$. The last two segments each map to a single abelian representation, an endpoint of B . On every C_J one has $|\tau| < 1$ (at the points with $v = \pi/2$ the limits are computed in Proposition 3.10).*

Proof. First let $\sin(mv) \sin(kv) \sin u \neq 0$. By Lemma 3.3 and the last statement of Proposition 3.4, the points of W are those with $v \in D_n$, $\cos u = \pm \frac{1}{2} \sqrt{1 - Q}$ and $\tau = \cot(mv) \cot u$. At an interior point $v \neq \pi/2$ of an interval of D_n , S is finite and nonzero and $\cos v S > 0$, so $\tau^2 < 1$ strictly by Lemma 3.3; at the folds $\tau = 0$. By (3.3), near a fold Q is a local coordinate, so the two branches over J join smoothly there, with u as local parameter, and C_J is a circle. It remains to treat the degenerate loci.

$\sin u = 0$. Then (E1) and (E2) read $\cos(mv) = 0$ and $\cos(m'v) = 0$. Writing $v = (2j + 1)\pi/(2m) = (2l + 1)\pi/(2m')$ and using $\gcd(m, m') = \gcd(m, \eta) = 1$, we get $m \mid 2j + 1$, so m is odd and $v = \pi/2$; then τ is arbitrary. These are the two segments in the statement. There $A_1 \mapsto \pm 1$, so the representation is abelian and does not depend on τ up to conjugation [HHK18, Thm. 10.2].

$\sin(mv) = 0$, $\sin u \neq 0$. Then (E1) gives $u = \pi/2$, and (E2) gives $\cos(m'v) = 0$ with $v = j\pi/m$. As before $m \mid 2j$, and $v \in \{0, \pi\}$ is excluded by $\cos(m'v) = 0$, so m is even and $v = \pi/2$. This is the fibre B , and conversely the fibre lies in W because $\sin(m\pi/2) = \cos(m'\pi/2) = 0$. In particular there are no solutions with $v \in \{0, \pi\}$.

$\sin(kv) = 0$, $\sin(mv) \sin u \neq 0$. By the computation in the proof of Lemma 3.3, the product $\cos(m'v) \sin(mv)$ vanishes, so $\cos(m'v) = 0$ with $v = j\pi/k$. Since $m' = 2k - \eta$, this forces $k \mid 2j$, hence k even (equivalently m odd) and $v = \pi/2$. Every u then solves, with $\tau = \cot(m\pi/2) \cot u = 0$. This is the line B . Conversely, the line lies in W , because both $\cos(m\pi/2)$ and $\cos(m'\pi/2)$ vanish.

Finally, for ν even the value $v = \pi/2$ is a zero of $\sin(mv)$ ($n \equiv 1$) or of $\sin(kv)$ ($n \equiv 2$), so it contributes only points of B . The statement about $|\tau|$ near $v = \pi/2$ on C_0 is part of Proposition 3.10. $\square$ $\square$

The endpoints of B are the two abelian traceless representations $r_{\pm}$ of [HHK18, §8.1]: the two ends of the fibre ($\tau = \pm 1$) in the fibre case, and the collapsed segments at $u \in \{0, \pi\}$ in the line case. Which of them is r_+ depends on $m \bmod 4$ and is determined in Proposition 3.9.

Lemma 3.7 (Symmetry). *The involution $\varsigma(u, v, \tau) = (u, \pi - v, -\tau)$ preserves W : it multiplies (E1) by $(-1)^m$ and (E2) by $(-1)^{m'}$. It exchanges C_l and C'_l and the halves C_0^L and C_0^R , and on images $\Pi \circ \varsigma = \Pi$ for n even and $\Pi \circ \varsigma = \iota \circ \Pi$ for n odd, where $\iota(\gamma, \theta) = (\pi - \gamma, -\theta)$.*

Proof. For the first claim use $\cos(j(\pi - v)) = (-1)^j \cos jv$ and $\sin(j(\pi - v)) = -(-1)^j \sin jv$. For the second, write $Q'_B = -\tau \mathbf{i} + \sqrt{1 - \tau^2} \mathbf{j}$ for the axis at $\varsigma(u, v, \tau)$. Then

$$e^{(\pi-v)Q'_B} = -e^{-vQ'_B} = -\mathbf{i} e^{vQ_B} \mathbf{i}^{-1},$$

while e^{uQ_A} commutes with $\mathbf{i}$. So the representation at $\varsigma(u, v, \tau)$ is conjugate to the one at (u, v, τ) multiplied by the central character χ with $\chi(A_1) = 1$, $\chi(B_1) = -1$. By (3.2), χ multiplies a, b, c by $(-1)^{q-r}, (-1)^r, (-1)^{q-r}$. If all three signs agree, the new triple is conjugate to the old one (by a rotation through π about the normal of their common plane), and the image is unchanged. If only b changes sign, γ becomes $\gamma + \pi$, which in the pillowcase is $(\pi - \gamma, -\theta)$; the case where only b keeps its sign is the same after the rotation. For $\eta = 1$ the signs are $((-1)^m, -1, (-1)^m)$, and for $\eta = -1$ they are $(-1, (-1)^m, -1)$. In both cases they agree exactly when m is odd, that is when n is even. $\square$ $\square$

Lemma 3.8 (Regular points). *The differential $d_{(u,v,\tau)}\Psi$ at $\epsilon = 0$ has rank 2 at every point of W except the two double points $c_{\pm}$ of Proposition 3.10 (when $n \equiv 4, 5 \pmod{6}$) and the collapsed segments at the ends of B in the line case. More precisely:*

- (1) *On a circle C_J , (E1) has $\partial_{\tau}(\text{E1}) = -\sin(mv) \sin u \neq 0$ away from $v = \pi/2$, and the remaining equation $G = -2 \cos^2 u/S - \cos(m'v)$ has $\partial_u G = 4 \sin u \cos u/S \neq 0$ off the folds and $\partial_v G = m' \sin(m'v_f) \neq 0$ at a fold.*
- (2) *On the line B (m odd), $\partial_u \Psi = 0$, and the (v, τ) -minor of $d\Psi$ is $\omega_1 \omega_2 \sin u (m' - 2k \cos^2 u)$ with $\omega_1 = \sin(m\pi/2)$, $\omega_2 = \sin(m'\pi/2)$.*
- (3) *On the fibre B (m even), $\partial_{\tau} \Psi = 0$, and the (u, v) -minor is $\omega_1 \omega_2 (2m\tau^2 - m')$ with $\omega_1 = \cos(m\pi/2)$, $\omega_2 = \sin(m'\pi/2)$.*

Proof. For (1), on C_J we have $\sin u \neq 0$ (as $x^2 < 1$) and $\sin(mv) \neq 0$ except at $v = \pi/2$ for $n \equiv 5$, so (E1) defines τ as a smooth function of (u, v) and the rank is 2 exactly when $dG \neq 0$. The formula for $\partial_u G$ vanishes only where $\cos u = 0$ (the folds) or $\sin(kv) = 0$. On a non-central circle $\sin(kv) \neq 0$ by the proof of Proposition 3.4. On C_0 the zeros of $\sin(mv)$ and $\sin(kv)$ nearest to $\pi/2$ other than $\pi/2$ itself are $\pi/2 \pm \pi/(2m)$ or $\pi/2 \pm \pi/m$, resp. $\pi/2 \pm \pi/k$ or $\pi/2 \pm \pi/(2k)$, all farther from $\pi/2$ than the half-width π/m' of J_0 ; at $v = \pi/2$ itself C_0 meets B (Proposition 3.10).

For (2), at $(u, \pi/2, 0)$ with m and m' odd, $\cos(m\pi/2) = \cos(m'\pi/2) = 0$, so

$$d(\text{E1}) = (0, -m\omega_1 \cos u, -\omega_1 \sin u), \quad d(\text{E2}) = (0, -m'\omega_2 \cos 2u, -\omega_2 \sin 2u),$$

whose (v, τ) -minor is $\omega_1 \omega_2 (m \cos u \sin 2u - m' \sin u \cos 2u) = \omega_1 \omega_2 \sin u (m' - 2k \cos^2 u)$, using $m' - m = k$. For (3), at $(\pi/2, \pi/2, \tau)$ with m even and m' odd,

$$d(\text{E1}) = (-\omega_1, -m\tau\omega_1, 0), \quad d(\text{E2}) = (2\tau\omega_2, m'\omega_2, 0),$$

with (u, v) -minor $\omega_1 \omega_2 (2m\tau^2 - m')$. $\square$ $\square$

For $n \equiv 2 \pmod{6}$ ($\eta = -1$, $k = m - 1$) the minor in (2) is $\omega_1 \omega_2 \sin u (2m - 1 - 2(m - 1) \cos^2 u)$, which vanishes only at $\sin u = 0$. For $n \equiv 1 \pmod{6}$ ($\eta = 1$, $m' = 2m + 1$) the minor in (3) never vanishes on $[-1, 1]$. For $n \equiv 4$ and $n \equiv 5$ the minors vanish exactly at the double points.

3.4. The binary dihedral arc in the pillowcase.

Proposition 3.9. *The image $\Pi(B)$ is the straight segment from the corner $(0, 0)$ to the corner $(\pi, 0)$ given, in a lift starting at the origin, by*

$n \bmod 6$	lift of $\Pi(B)$	γ on B	$\varphi = \theta - \gamma$ on the lift
1	$t \mapsto (t, 2t)$	$\arccos \tau \ (m \equiv 0), \ \pi - \arccos \tau \ (m \equiv 2)$	t
2	$t \mapsto (t, 4t)$	$\pi - u \ (m \equiv 1), \ u \ (m \equiv 3)$	$3t$
4	$t \mapsto (t, -2t)$	$u \ (m \equiv 1), \ \pi - u \ (m \equiv 3)$	$-3t$
5	$t \mapsto (t, 0)$	$\pi - \arccos \tau \ (m \equiv 0), \ \arccos \tau \ (m \equiv 2)$	$-t$

with $t \in [0, \pi]$ and m taken modulo 4. The endpoint of B mapping to $(0, 0)$ is r_+ , and the one mapping to $(\pi, 0)$ is r_- . The slopes agree with [FKP17, Thm. 4.1].

Proof. At $v = \pi/2$ we have $\rho_B = e^{(\pi/2)Q_B} = Q_B$, so ρ_B^j is ± 1 or $\pm Q_B$ according to $j \bmod 4$, and (3.2) can be evaluated directly. We identify traceless unit quaternions with unit vectors.

$n \equiv 4$ ($\eta = 1$, m odd, $m' \equiv 3 \bmod 4$). Here $Q_B = \mathbf{j}$, $\rho_A = e^{u\mathbf{i}}$ and $a = \rho_A \rho_B^m = \alpha e^{u\mathbf{i}} \mathbf{j}$ with $\alpha = (-1)^{(m-1)/2}$, $b = \rho_B^{-m'} \rho_A^{-2} = \mathbf{j} e^{-2u\mathbf{i}}$ and $c = \rho_B^{-m'} a \rho_B^{m'} = -\mathbf{j} a \mathbf{j}$. In the $\mathbf{jk}$ -plane, with angles measured from $\mathbf{j}$ towards $\mathbf{k}$, the vectors a, b, c have angles $u, 2u, -u$ if $\alpha = 1$, and $u + \pi, 2u, \pi - u$ if $\alpha = -1$. In the first case the oriented angle from a to b is $u \in (0, \pi)$, so $\gamma = u$ and $\theta = -2u$. In the second it is $u - \pi$, so $\gamma = \pi - u$, the orientation of the frame is reversed, and $\theta = 2u \equiv -2\gamma$.

$n \equiv 2$ ($\eta = -1$, m odd, $m' \equiv 1 \bmod 4$). Now $a = e^{2u\mathbf{i}} \mathbf{j}$ (angle $2u$), $c = -\mathbf{j} a \mathbf{j}$ (angle $-2u$), and $b = \mathbf{j}^{-m} e^{-u\mathbf{i}}$ has angle $u + \pi$ if $m \equiv 1$ and u if $m \equiv 3 \pmod{4}$. This gives $(\gamma, \theta) = (\pi - u, -4u) \equiv (\gamma, 4\gamma)$, resp. $(u, 4u)$.

$n \equiv 1$ ($\eta = 1$, m even, $m' \equiv 1 \bmod 4$). Here $u = \pi/2$, $\rho_A = \mathbf{i}$ and $\rho_B = Q_B$, which makes the angle $\beta = \arccos \tau$ with $\mathbf{i}$ in the $\mathbf{ij}$ -plane. Then $a = (-1)^{m/2} \mathbf{i}$, $b = Q_B$ and $c = -Q_B a Q_B$, which is a reflected in the line of Q_B . So a, b, c have angles $0, \beta, 2\beta$ if $m \equiv 0$, and $\pi, \beta, 2\beta + \pi$ if $m \equiv 2 \pmod{4}$, giving $(\gamma, \theta) = (\beta, 2\beta)$, resp. $(\pi - \beta, -2\beta) \equiv (\gamma, 2\gamma)$.

$n \equiv 5$ ($\eta = -1$, m even, $m' \equiv 3 \bmod 4$). Here $a = -\rho_B^{m'} = Q_B$, $b = \rho_B^{-m} \rho_A^{-1} = -(-1)^{m/2} \mathbf{i}$ and $c = \rho_B^{-m} a \rho_B^m = a$. Thus $\theta = 0$ and $\cos \gamma = a \cdot b = -(-1)^{m/2} \tau$.

In every case the endpoints of B go to $\gamma \in \{0, \pi\}$ and $\theta \in 2\pi\mathbb{Z}$, that is, to the corners $(0, 0)$ and $(\pi, 0)$. Following [HHK18, §§5.3, 6.1], the abelian representation mapping to $(0, 0)$ is denoted r_+ . □

3.5. The central circle and its double points.

Proposition 3.10. (1) *If $n \equiv 1, 2 \pmod{6}$, then ν is even, W is the disjoint union of B and the ν non-central circles, and $d\Psi$ has rank 2 on W away from the two ends of B .*

(2) *If $n \equiv 4, 5 \pmod{6}$, then ν is odd, C_0 is a smooth circle, the non-central circles are disjoint from $B \cup C_0$, and $C_0 \cap B$ consists of two points $c_\pm$:*

$n \bmod 6$	$\{c_+, c_-\}$	$\{\Pi(c_+), \Pi(c_-)\}$
4	$(u_1, \frac{\pi}{2}, 0), (\pi - u_1, \frac{\pi}{2}, 0), \cos^2 u_1 = \frac{2m+1}{2m+2}, u_1 \in (0, \frac{\pi}{2})$	$(\gamma_*, -2\gamma_*), (\pi - \gamma_*, 2\gamma_*)$
5	$(\frac{\pi}{2}, \frac{\pi}{2}, \pm\tau_1), \tau_1 = \sqrt{1 - \frac{1}{2m}}$	$(\gamma_*, 0), (\pi - \gamma_*, 0)$

Here $\gamma_* = u_1$ for $n \equiv 4$ and $\cos \gamma_* = \tau_1$ for $n \equiv 5$, so $\cos^2 \gamma_* = \frac{2m+1}{2m+2}$, resp. $1 - \frac{1}{2m}$, and $0 < \gamma_* \leq \pi/6$, with equality exactly for $n = 4, 5$. Each half C_0^L, C_0^R is an arc from c_+ to c_- through a fold, and ς exchanges them.

Proof. If $n = 3m + \eta$, then $\nu = m$ for $\eta = 1$ and $\nu = m - 1$ for $\eta = -1$, so ν is even for $n \equiv 1, 2$ and odd for $n \equiv 4, 5 \pmod{6}$. By Proposition 3.4, the non-central intervals stay at distance at least $2\pi/N$ from $\pi/2$, so the non-central circles are disjoint from $B \subset \{v = \pi/2\}$.

For (1), W has no central circle, so by Proposition 3.6 it is B together with the non-central circles, and these are disjoint. The rank statement is Lemma 3.8 together with the remark after it. In the main branch near $v = \pi/2$ the solutions would have $\tau^2 \rightarrow (2m+1)/(2m) > 1$ ($n \equiv 1$) or $x^2 > 1$ ($n \equiv 2$), in agreement with $\pi/2 \notin D_n$.

For (2), consider C_0 near $v = \pi/2$ and write $v = \pi/2 + w$.

If $n \equiv 4$, then m is odd and n, k are even with $(n-k)/2 = m$ odd, so Q extends smoothly across $v = \pi/2$ with $Q(\pi/2) = -n/k$. Hence $4x^2 \rightarrow 1 + n/k = N/k$, that is $\cos^2 u \rightarrow m'/(2k) = (2m+1)/(2m+2)$, and $\tau = \cot(mv) \cot u \rightarrow 0$ because $\cot(m\pi/2) = 0$. Each branch of C_0 passes smoothly through $v = \pi/2$, at $(u_1, \pi/2, 0)$ on the branch $x > 0$ and at $(\pi - u_1, \pi/2, 0)$ on the branch $x < 0$.

If $n \equiv 5$, then m is even and n, k are odd with $(n-k)/2 = m$ even, and $Q(\pi/2 + w) = \cos(nw)/\cos(kw) = 1 - \frac{n^2-k^2}{2}w^2 + O(w^4)$. The solutions of $4x^2 = 1 - Q$ near $w = 0$ are the two smooth curves $x = \pm \left(\frac{n^2-k^2}{8}\right)^{1/2}w + O(w^3)$, along which $u \rightarrow \pi/2$ and

$$\tau = \cot(mv) \cot u \rightarrow \pm \frac{1}{m} \left(\frac{n^2-k^2}{8} \right)^{1/2} = \pm \tau_1, \quad \tau_1^2 = \frac{2m(4m-2)}{8m^2} = 1 - \frac{1}{2m},$$

on the curve with leading coefficient $\pm$, using $\cot u = x/\sqrt{1-x^2}$, $n-k = 2m$, $n+k = 4m-2$ and $\cot(m(\pi/2+w)) = \cot(mw)$. So C_0 passes through $(\pi/2, \pi/2, \pm\tau_1)$, which lie on the fibre B .

In both cases these are the only points of C_0 with $v = \pi/2$, and they are the only points of $C_0 \cap B$. The points $c_{\pm}$ lie on the two branches. Following C_0 from c_+ through $v < \pi/2$, one reaches the fold at $(2\nu-1)\pi/N$ and returns along the other branch to c_- ; this is C_0^L , and $C_0^R = \varsigma(C_0^L)$ by Lemma 3.7. Smoothness of C_0 as a curve follows from the explicit parametrizations above, from (3.3) at the folds, and from Lemma 3.8(1) elsewhere. The images follow from Proposition 3.9: for $n \equiv 4$ the points u_1 and $\pi - u_1$ of the line go to $\gamma \in \{u_1, \pi - u_1\}$ on the segment $\theta \equiv -2\gamma$, and for $n \equiv 5$ the points $\tau = \pm\tau_1$ of the fibre go to $\gamma \in \{\gamma_*, \pi - \gamma_*\}$ on $\theta = 0$. Finally $\cos^2 \gamma_* \geq 3/4$, with equality for $m = 1$ ($n = 4$) and $m = 2$ ($n = 5$), and it increases with m . $\square$ $\square$

Proposition 3.11 (Nondegenerate double points). *Let $n \equiv 4, 5 \pmod{6}$ and let c be one of $c_{\pm}$. Then $d_{(u,v,\tau)}\Psi(c)$ has rank 1, its kernel is spanned by the tangent vectors T_B of B and T_C of C_0 at c , and the reduced Hessian $\ell \cdot D^2\Psi(c)$ restricted to the kernel, with ℓ spanning the left kernel of $d\Psi(c)$, has the form $\begin{pmatrix} 0 & \Lambda \\ \Lambda & 0 \end{pmatrix}$ in the basis (T_B, T_C) , with $\Lambda \neq 0$. Explicitly, with respect to the equations (E1, E2):*

- (1) for $n \equiv 4$ and $c = (u_1, \pi/2, 0)$: $T_B = \partial_u$ and $T_C = \sin u_1 \partial_v - m \cos u_1 \partial_\tau$; moreover $\ell = (\omega_2 \sin 2u_1, -\omega_1 \sin u_1)$, and

$$\Lambda = \ell \cdot D^2\Psi(T_B, T_C) = -2(m+1)\omega_1\omega_2 \sin 2u_1 \sin^2 u_1;$$

- (2) for $n \equiv 5$ and $c = (\pi/2, \pi/2, \tau)$, $\tau = \pm\tau_1$: $T_B = \partial_\tau$, the kernel is spanned by T_B and $X = -\tau m \partial_u + \partial_v$, $\ell = (2\tau\omega_2, \omega_1)$, and

$$\ell \cdot D^2\Psi(T_B, X) = -4\tau m \omega_1 \omega_2,$$

with ω_1, ω_2 as in Lemma 3.8. In particular c is a nondegenerate (Morse) double point of W : near c , W is the union of two smooth curves meeting transversally.

Proof. For (1), the differentials of (E1) and (E2) at c are given in the proof of Lemma 3.8; their u -components vanish, and they are proportional exactly when $m \cos u \sin 2u = m' \sin u \cos 2u$, that is $\cos^2 u = m'/(2k) = (2m+1)/(2m+2)$. This identifies $c_{\pm}$ as the points of B where the rank drops, and gives $\ker d\Psi(c) = \langle \partial_u, \sin u_1 \partial_v - m \cos u_1 \partial_\tau \rangle$ and the stated ℓ . The mixed second derivatives at c are

$$\begin{aligned} \partial_u \partial_v(\text{E1}) &= m\omega_1 \sin u_1, & \partial_u \partial_\tau(\text{E1}) &= -\omega_1 \cos u_1, \\ \partial_u \partial_v(\text{E2}) &= 2m'\omega_2 \sin 2u_1, & \partial_u \partial_\tau(\text{E2}) &= -2\omega_2 \cos 2u_1, \end{aligned}$$

so $D^2(\text{E1})(T_B, T_C) = m\omega_1$ and

$$D^2(\text{E2})(T_B, T_C) = 2\omega_2(m' \sin u_1 \sin 2u_1 + m \cos u_1 \cos 2u_1).$$

Hence

$$\Lambda = \omega_1 \omega_2 \sin 2u_1 (m - 2m' \sin^2 u_1 - m \cos 2u_1) = 2\omega_1 \omega_2 \sin 2u_1 \sin^2 u_1 (m - m'),$$

and $m - m' = -(m + 1)$. The tangent of C_0 at c has zero u -component, because $Q(\pi - v) = Q(v)$ makes x an even function of $v - \pi/2$ along C_0 ; as it lies in the kernel, it is a multiple of T_C . Since $B \subset W$ and $C_0 \subset W$ pass through c with tangents T_B and T_C , the reduced Hessian vanishes on each of them, so it is $\begin{pmatrix} 0 & \Lambda \\ \Lambda & 0 \end{pmatrix}$.

For (2), the differentials are given in the proof of Lemma 3.8(3). They are proportional exactly when $2m\tau^2 = m'$, i.e. $\tau^2 = 1 - \frac{1}{2m}$, and then $d(\text{E2}) = -2\tau\omega_1\omega_2 d(\text{E1})$, which gives ℓ and the kernel. The only nonzero mixed derivatives involving ∂_τ are $\partial_\tau \partial_v(\text{E1}) = -m\omega_1$ and $\partial_\tau \partial_u(\text{E2}) = 2\omega_2$, so $\ell \cdot D^2\Psi(T_B, X) = 2\tau\omega_2(-m\omega_1) + \omega_1(-2\tau m\omega_2) = -4\tau m\omega_1\omega_2$. Also $\ell \cdot D^2\Psi(T_B, T_B) = 0$ since Ψ is affine in τ . So in the basis (T_B, X) the reduced Hessian is $\begin{pmatrix} 0 & \Lambda \\ \Lambda & * \end{pmatrix}$ with $\Lambda = -4\tau m\omega_1\omega_2 \neq 0$, which is nondegenerate and indefinite; its second null direction is the tangent of C_0 . $\square \quad \square$

Remark 3.12. For $n \equiv 4 \pmod{6}$ the line $B = \{v = \pi/2, \tau = 0\}$ lies in $W_{(\epsilon_A, 0)}$ for every ϵ_A . Indeed, with $\epsilon_B = 0$ the phases $\Theta_1 = mv$ and $\Theta_2 = -m'v$ give $\cos \Theta_1 = \cos \Theta_2 = 0$ at $v = \pi/2$, and $\tau = 0$ kills the second terms of (3.1); the same formulas give $\partial_u \Psi = 0$ along B . Hence at points of B the rank of $d\Psi$ is the rank of its (v, τ) -minor. At $\epsilon = 0$ this minor is $\omega_1 \omega_2 \sin u (m' - 2k \cos^2 u)$ (Lemma 3.8), which changes sign transversally at the parameters $u_1, \pi - u_1$ of $c_\pm$. By continuity it still vanishes at some point of B near each of them for every small ϵ_A , and there $d\Psi$ has rank at most one. So perturbations with $\epsilon_B = 0$ never resolve $c_\pm$ for these knots. They are excluded by the hypothesis $\epsilon \in \mathcal{C}_a$ of Theorem 1.1 (Definition 5.5). In [HHK18, §11.6] the perturbation for $T(3, 4)$ is taken with $\epsilon_B = 0$, and $R_{\epsilon_A, 0}(Y, T)$ is stated there to be the union of an arc and a circle for every small nonzero ϵ_A . For small $\epsilon \in \mathcal{C}_a$, which forces $\epsilon_B \neq 0$, our computations give the split resolution of Section 6, that is, the arc and circle drawn in [HHK18, Figs. 20–21], and Theorem 6.10 then gives the complex computed there.

4. MONOTONICITY OF THE ANGLE φ

Throughout this section $\epsilon_A = \epsilon_B = 0$, and $\Pi: W \rightarrow P$ is the restriction map of the unperturbed tangle. On a lift of Π to $\mathbb{R}^2$ we use the function $\varphi = \theta - \gamma$; the preimage of the diagonal arc Δ is $\bigcup_k \Delta_k$ with $\Delta_k = \{\varphi = 2\pi k\}$. The main results are Theorem 4.4, which says that φ is strictly monotone along every non-central circle, and Theorem 4.10, which says the same for each half of the central circle. Together with Propositions 4.11 and 4.12 and Lemma 4.13, they contain all the information about the unperturbed curves that Sections 5 and 6 use. Three inequalities between explicit trigonometric functions of two variables enter the proofs; they are verified by rigorous interval-arithmetic computations in Appendix A.

4.1. Closed forms. Recall from Section 3 that on the part of W where $\sin u \sin(mv) \sin(kv) \neq 0$ the equations (E1), (E2) are equivalent to

$$4x^2 = 1 - Q(v), \quad \tau = \cot(mv) \cot u, \quad x = \cos u,$$

so that this part of W is a double cover of its projection to the v -axis, by the two branches $x > 0$ and $x < 0$. We write

$$\psi = mv, \quad \delta = \eta v, \quad a = \psi, \quad b = \psi + \delta = kv,$$

so that $a + b = m'v$ and $2a + b = nv$, and we put

$$A = -\cos(a + b), \quad B = \cos \delta, \quad C = \sin^2 a + \sin^2 b, \quad s_3 = \sin(2a + b),$$

$$P_* = 2 \cos a - \cos 3a - \cos(a + 2b).$$

Let $M = \rho(A_1) = e^{uQ_A}$ and $N = \rho(B_1) = e^{vQ_B}$ be the images of the two generators of $\pi_1(Y \setminus T)$ used in Section 3, so that $Q_A \cdot Q_B = \tau$.

Lemma 4.1. *On the part of W where $\sin u \sin(mv) \sin(kv) \neq 0$,*

$$\cos \gamma = \eta x \frac{\sin v}{\sin(mv)}, \quad (4.1)$$

$$\cos \varphi = \cos 3u \cos nv - \tau \sin 3u \sin nv = x \frac{P_*}{2 \sin a \sin b}, \quad (4.2)$$

$$x^2 = \frac{A \sin a}{2 \sin b}, \quad \cos^2 \varphi = \frac{A P_*^2}{8 \sin a \sin^3 b}, \quad \sin^2 \varphi = \frac{s_3^2 B C}{2 \sin a \sin^3 b}. \quad (4.3)$$

Proof. With $\epsilon_A = \epsilon_B = 0$ the elements A_2, B_2 of HHK2 (31) are trivial, so $a = A_1^{s+p} B_1^{q-r}$, $b = B_1^{-r} A_1^s$ and $c = B_1^{-r} a B_1^r$. In the normalization $a \mapsto \mathbf{i}$, $b \mapsto e^{\gamma \mathbf{k} \mathbf{i}}$, $c \mapsto e^{\theta \mathbf{k} \mathbf{i}}$ that defines the pillowcase coordinates (Section 2) one has $\rho(c)\rho(b)^{-1} = e^{(\theta-\gamma)\mathbf{k}}$, so $\cos \varphi = \operatorname{Re} \rho(cb^{-1})$. Now

$$cb^{-1} = B_1^{-r} A_1^{s+p} B_1^q A_1^{-s} B_1^r,$$

which is conjugate to $A_1^p B_1^q$. Since $\operatorname{Re}(e^{\alpha Q_A} e^{\beta Q_B}) = \cos \alpha \cos \beta - \tau \sin \alpha \sin \beta$, this gives the first expression in (4.2). Similarly $\cos \gamma = -\operatorname{Re} \rho(ab)$, and ab is conjugate to $A_1^{2s+p} B_1^{q-2r} = A_1^{-\eta} B_1^{-\eta k}$, so $\cos \gamma = -\cos u \cos kv + \tau \sin u \sin kv$; substituting $\tau \sin u = \cot(mv) \cos u$ and $\sin(kv - mv) = \sin(\eta v)$ gives (4.1).

For the second expression in (4.2), write $\sin 3u = \sin u (4x^2 - 1) = -Q \sin u$ and $\cos 3u = x(4x^2 - 3) = -(Q + 2)x$, and substitute $\tau \sin u = \cot(mv) x$. This gives $\cos \varphi = x Y(v)$ with $Y = -(Q + 2) \cos(2a + b) + Q \cot a \sin(2a + b)$. The identities

$$\frac{1}{4}(1 - Q) = \frac{A \sin a}{2 \sin b}, \quad Y = \frac{P_*}{2 \sin a \sin b}, \quad 8 \sin a \sin^3 b - A P_*^2 = 4 s_3^2 B C$$

are identities between trigonometric polynomials in a and b (after clearing denominators); each is checked by expanding both sides as Laurent polynomials in e^{ia} and e^{ib} . The first is x^2 , the product of the first two gives $\cos^2 \varphi$, and the third gives $\sin^2 \varphi = 1 - \cos^2 \varphi$. $\square$

Along either branch, v is a local parameter away from the folds, and $\delta = s\psi$ with $s = \eta/m$. Define

$$E = \cos \delta F_0 + s F_1, \quad F_0 = 10 \sin^2 \psi + 6 \sin^2(\psi + \delta) - 4 \sin^2(2\psi + \delta), \\ F_1 = 4 \cos \delta \sin^2 \psi - 2 \sin \delta \sin \psi \cos(3\psi + 2\delta).$$

Lemma 4.2. *Along either branch, at every point where $ABC \sin a \sin b \neq 0$,*

$$\frac{d}{d\psi} \cos^2 \varphi = \frac{E P_* s_3}{8 \sin^2 a \sin^4 b}, \quad \text{hence} \quad \left(\frac{d\varphi}{d\psi} \right)^2 = \frac{E^2}{16 A B C \sin^2 a \sin^2 b}. \quad (4.4)$$

Proof. Differentiate the expression for $\cos^2 \varphi$ in (4.3) along $\delta' = s$. The resulting identity is again an identity of trigonometric polynomials in ψ, δ , separately in the parts of degree 0 and 1 in s , and is checked by expansion in $e^{i\psi}, e^{i\delta}$. Where $P_* s_3 \neq 0$, combine it with $(d \cos^2 \varphi / d\psi)^2 = 4 \cos^2 \varphi \sin^2 \varphi (d\varphi / d\psi)^2$ and (4.3):

$$\left(\frac{d\varphi}{d\psi} \right)^2 = \frac{E^2 P_*^2 s_3^2}{64 \sin^4 a \sin^8 b} \cdot \frac{16 \sin^2 a \sin^6 b}{4 A B C P_*^2 s_3^2} = \frac{E^2}{16 A B C \sin^2 a \sin^2 b}.$$

Both sides are continuous on the branch, and the zeros of $P_* s_3$ are isolated, so the identity holds wherever $ABC \sin a \sin b \neq 0$. $\square$

So on a branch, φ has a critical point exactly where $E = 0$, apart from the points where $ABC \sin a \sin b = 0$. At a fold the branch is not parametrized by v , and we use the following direct computation.

Lemma 4.3. *At a fold of W , that is a point with $x = 0$ and $\cos(m'v) = 0$, the parameter u is a local coordinate on W and*

$$\left| \frac{d\varphi}{du} \right| = \frac{1}{|\sin a|} + 2|\sin a| \geq 2\sqrt{2}.$$

Proof. That u is a local coordinate follows from (3.3). Differentiating $4\cos^2 u = 1 - Q(v)$ shows $dv/du = 0$ at $u = \pi/2$. Differentiating (4.2) then gives $\sin \varphi \cdot d\varphi/du = \sin u \cdot P_*/(2\sin a \sin b)$ there, with $\sin \varphi = \pm 1$ and $\sin u = 1$. At a fold $a + b \equiv \pi/2 \pmod{\pi}$, so $\sin b = \pm \cos a$ and $\cos(a + 2b) = -\cos a$, hence $P_* = 3\cos a - \cos 3a = 2\cos a(1 + 2\sin^2 a)$ and $|P_*/(2\sin a \sin b)| = 1/|\sin a| + 2|\sin a|$. The bound is the arithmetic–geometric mean inequality. $\square$

4.2. The non-central circles. Let C be a non-central circle, over an interval $J_l = [(4l - 3)\pi/N, (4l - 1)\pi/N] \subset (0, \pi/2)$ or over its mirror $\pi - J_l$; here $1 \leq l \leq \lfloor \nu/2 \rfloor$ (Proposition 3.4). The two branches $x > 0$ and $x < 0$ over the interior of the interval are joined at two folds over its endpoints. Put

$$\Omega_+ = \{(\psi, \delta) : 0 \leq \delta \leq \frac{\pi}{2}, \frac{\pi}{4} - \frac{\delta}{2} \leq \psi \leq \frac{3\pi}{4} - \frac{\delta}{2}\}.$$

Theorem 4.4. *Along every non-central circle, φ is strictly monotone: in a regular parametrization of the circle, the derivative of any continuous lift of φ never vanishes.*

Proof. Circles over $J_l \subset (0, \pi/2)$. On the interior of J_l we have $A = -\cos(m'v) > 0$ (Proposition 3.4), $B = \cos v > 0$, $C > 0$, and $\sin a \sin b = \sin(mv) \sin(kv) \neq 0$. By Lemma 4.2 it suffices to show $E \neq 0$ there; the folds are covered by Lemma 4.3.

The functions F_0 and F_1 , hence E , are π -periodic in ψ . On J_l we have $2\psi + \delta = m'v \in [\pi/2, 3\pi/2] + 2\pi(l - 1)$, because $N = 2m'$; hence $\psi - (l - 1)\pi$ lies in $[\pi/4 - \delta/2, 3\pi/4 - \delta/2]$, and $|\delta| = v < \pi/2$. So the reduced point lies in Ω_+ if $\eta = 1$ and in $\Omega_- = \{-\frac{\pi}{2} \leq \delta \leq 0, \frac{\pi}{4} - \frac{\delta}{2} \leq \psi \leq \frac{3\pi}{4} - \frac{\delta}{2}\}$ if $\eta = -1$. The map $(\psi, \delta) \mapsto (\pi - \psi, -\delta)$ carries Ω_- onto Ω_+ , and it leaves F_0 and F_1 unchanged, since both are even under $(\psi, \delta) \mapsto (-\psi, -\delta)$ and π -periodic in ψ . Write $E/\cos \delta = F_0 + \lambda F_1$ with $\lambda = s/\cos \delta$. The right endpoint of J_l is at most $\pi/2 - 2\pi/N$, because $4l - 1 \leq 2\nu - 1$, so $\cos \delta = \cos v \geq \sin(\pi/m') \geq \sin(\pi/(2m + 1))$ and

$$|\lambda| \leq \frac{1}{m \sin(\pi/(2m + 1))} \leq \frac{1}{2 \sin(\pi/5)} = 0.85065 \dots$$

The middle inequality holds because $m \sin(\pi/(2m + 1)) = \frac{\pi m}{2m + 1} \operatorname{sinc} \frac{\pi}{2m + 1}$, where $\operatorname{sinc} y = \sin y/y$, is a product of two increasing functions of m , and non-central circles exist only for $\nu \geq 2$, that is $m \geq 2$. (The extreme value is attained for $n = 7$.) Hence

$$E/\cos \delta \geq F_0 - 0.851 |F_1| \geq 1.648 \quad \text{on } J_l,$$

by Lemma A.1. So $E > 0$ on the interior of J_l .

Mirror circles. The involution $\varsigma(u, v, \tau) = (u, \pi - v, -\tau)$ of Lemma 3.7 multiplies (E1) by $(-1)^m$ and (E2) by $(-1)^{m'}$, so it preserves W and carries the circle over J_l onto the circle over $\pi - J_l$. By Lemma 4.7 below, $\varphi \circ \varsigma \equiv \varphi$ if n is even and $\varphi \circ \varsigma \equiv -\varphi - \pi$ if n is odd, modulo 2π . Along a circle, the difference between a continuous lift of $\varphi \circ \varsigma$ and the corresponding continuous lift of φ , resp. $-\varphi - \pi$, is continuous with values in $2\pi\mathbb{Z}$, hence constant, and strict monotonicity carries over. $\square$

The following facts about γ and the diagonal are consequences.

Proposition 4.5. *Every non-central circle, and the open halves of the central circle, satisfy $|\cos \gamma| < 1$. In particular they miss the side edges $\gamma \in \{0, \pi\}$ of the pillowcase, and the corners.*

Proof. For $v \in (0, \pi/2)$, square (4.1). By Lemma 3.2, $x^2 = -\cos(m'v)S(v)/2$ with $S = \sin(mv)/\sin(kv)$, and $2\sin(mv)\sin(kv) = \cos v - \cos(m'v)$. Hence $x^2 \sin^2 v < \sin^2(mv)$ is equivalent to $\cos(m'v)\cos^2 v < \cos v$. On $D_n \cap (0, \pi/2)$ we have $\cos(m'v) \leq 0 < \cos v$, so the inequality holds. For $v \in (\pi/2, \pi)$ apply ς , which acts on the pillowcase as the identity if n is even and as $\iota: (\gamma, \theta) \mapsto (\pi - \gamma, -\theta)$ if n is odd (Lemma 4.7 below); both preserve $|\cos \gamma|$. $\square$

Proposition 4.6. *Let C be a non-central circle. Exactly four points of C satisfy $\varphi \in \pi\mathbb{Z}$, two with $\varphi \equiv 0$ and two with $\varphi \equiv \pi \pmod{2\pi}$, and they alternate around C . Consequently a lift of φ changes by $\pm 4\pi$ once around C , the image of C meets Δ in exactly two points, transversally, and C has vertical degree 2: its lift to $\mathbb{R}^2$ runs from a point z to $z \pm (0, 4\pi)$ inside the strip $0 < \gamma < \pi$.*

Proof. By (4.2), $\varphi \in \pi\mathbb{Z}$ if and only if $\rho(A_1)^3 \rho(B_1)^n = \pm 1$. Since $|\tau| < 1$ on C , the axes Q_A and Q_B are not parallel, so this forces $\sin 3u = 0$ and $\sin nv = 0$; then $Q(v) = 0$ and $x = \pm \frac{1}{2}$. Conversely every zero of $\sin nv$ in the interior of the interval gives two such points, one on each branch.

For $J_l \subset (0, \pi/2)$ the zeros are $v_j = j\pi/n$ with $(4l-3)n/N \leq j \leq (4l-1)n/N$. Since $n/N = \frac{3}{4} - \frac{\eta}{4m'}$ and $4l-1 \leq 2\nu-1 < m'$, we have $(4l-3)n/N \in (3l - \frac{5}{2}, 3l-2)$ and $(4l-1)n/N \in (3l-1, 3l - \frac{1}{2})$, so the zeros are exactly $j = 3l-2$ and $j = 3l-1$. At v_j we have $\cos \varphi = \cos 3u \cos(j\pi)$, and $\cos 3u = -1$ for $x = \frac{1}{2}$, $\cos 3u = 1$ for $x = -\frac{1}{2}$. Thus on the branch $x > 0$, $\varphi \equiv 0$ if and only if j is odd, and on the branch $x < 0$ if and only if j is even. Going once around C (branch $x > 0$ from v_{3l-2} to v_{3l-1} , through a fold, then back along $x < 0$) the labels of the four points therefore alternate between 0 and π .

By Theorem 4.4, a lift of φ is strictly monotone. Between two consecutive points of the list it cannot meet a multiple of π , so it changes by exactly $\pm\pi$; the total change is $\pm 4\pi$, and the two points with $\varphi \equiv 0$ are transverse intersections with Δ . By Proposition 4.5 the lift of γ stays in $(0, \pi)$, so $\theta = \varphi + \gamma$ changes by $\pm 4\pi$ as well. The mirror circles follow by Lemma 4.7. $\square$

Lemma 4.7. *The involution ς is induced by multiplying a representation by the central character χ of $\pi_1(Y \setminus T)$ with $\chi(A_1) = 1$ and $\chi(B_1) = -1$. On the pillowcase it acts as the identity if n is even and as $\iota: (\gamma, \theta) \mapsto (\pi - \gamma, -\theta)$ if n is odd. In particular it maps φ to φ , resp. to $-\varphi - \pi$, modulo 2π .*

Proof. $-e^{vQ_B} = e^{(\pi-v)(-Q_B)}$, and replacing Q_B by $-Q_B$ replaces τ by $-\tau$. By HHK2 (31), $\chi(a) = (-1)^{q-r}$ and $\chi(b) = (-1)^r$, and c is conjugate to a . For $\eta = 1$, $(q-r, r) = (m, m')$, and for $\eta = -1$, $(q-r, r) = (m', m)$; here m' is odd and $m \equiv n+1 \pmod{2}$. If n is even, a , b and c are therefore all multiplied by -1 , which does not change their pillowcase coordinates. If n is odd, either b alone or a and c together change sign. After rotating $-a$ back to $\mathbf{i}$ in the second case, both cases replace the angles (γ, θ) of b and c by $(\gamma + \pi, \theta)$, and restoring the normalization ($\gamma \in [0, \pi]$, by reversing the orientation of the plane) gives $(\pi - \gamma, -\theta)$. $\square$

Corollary 4.8. *Every non-central circle C is a restricted immersed circle in the sense of HHK2 Def. 3.6 that misses the left and right edges of the pillowcase. Its class in $\pi_1(P^*)$ is the square of the class of the core of the annulus $\{0 < \gamma < \pi\} \subset P^*$. For small $\epsilon > 0$ it meets L_0 in exactly four points, and it contributes $\mathbb{Z}/4$ -graded rank $(1, 1, 1, 1)$ to $H^\natural$, with no differential. All of this persists under C^1 -small perturbations of C that keep the derivative of φ nonzero.*

Proof. By Theorem 4.4 the velocity of the image is never parallel to $(1, 1)$, so $L_1|_C$ is an immersion and $\mu(L_1|_C, \ell_1) = 0$. By Proposition 4.6 the image lies in the annulus $\{0 < \gamma < \pi\}$ and winds twice around it, so its class is $c^{\pm 2}$ for the core c , which is nontrivial in the free group $\pi_1(P^*)$.

The core encircles two corners, so $z(c) = \pm 2$ and $z(C) = \pm 4 \equiv 0$; hence $\mu + z \equiv 0 \pmod{4}$. An immersed monogon would be bounded by a sub-arc of C whose two ends have the same image and whose closed-up image is null-homotopic in P^* . Such a loop lifts to a closed loop in $\mathbb{R}^2 \setminus (\pi\mathbb{Z})^2$, along which a lift of φ returns to its initial value; this contradicts strict monotonicity. So C is unobstructed (HHK2 Def. 2.1), hence restricted. HHK2 Thm. 5.4 then gives the graded rank $(d/2)^{\times 4}$ with vertical degree $d = 2$. For small ϵ , L_0 lies within $O(\epsilon)$ of Δ away from the corners, and C misses the corners, so the intersection points of L_0 with C are the two points $x^\pm$ near each of the two transverse intersections with Δ . Four generators of total rank four leave no room for a differential. The last sentence holds because every property used is open in the C^1 topology, or is homotopy invariant. $\square$

4.3. The central circle. Now let $n \equiv 4, 5 \pmod{6}$, so that ν is odd and W contains the central circle C_0 over $J_0 = [v_L, \pi - v_L]$, $v_L = (2\nu - 1)\pi/N$. It meets B at the two points of Proposition 3.10. We write c_+ for the crossing whose image has $\gamma = \gamma_*$, and c_- for the crossing whose image has $\gamma = \pi - \gamma_*$. The left half C_0^L (where $v \leq \pi/2$) is an arc from one crossing to the other: it starts at $v = \pi/2$ on the branch $x > 0$, runs down to the fold over v_L , and returns to $v = \pi/2$ along $x < 0$. The right half is $C_0^R = \varsigma(C_0^L)$. On the left half put

$$v = \frac{\pi}{2} - w, \quad 0 < w \leq \frac{2\pi}{N}, \quad \xi = mw, \quad h = \frac{1}{m}.$$

Then $\xi \in (0, \pi/(2+h)]$ with $h \in (0, 1]$ for $n \equiv 4$ (where m is odd), and $\xi \in (0, \pi/(2-h)]$ with $h \in (0, \frac{1}{2}]$ for $n \equiv 5$ (where m is even). Here and below $n \equiv 4$ and $n \equiv 5$ mean residues modulo 6.

Lemma 4.9. *On the left half, $E = h \mathcal{E}_n(\xi, h)$, where for $n \equiv 4$*

$$\begin{aligned} \mathcal{E}_4 = \frac{\sin(\xi h)}{h} [10 \cos^2 \xi + 6 \sin^2(\xi + \xi h) - 4 \cos^2(2\xi + \xi h)] \\ + 4 \sin(\xi h) \cos^2 \xi - 2 \cos(\xi h) \cos \xi \sin(3\xi + 2\xi h), \end{aligned}$$

and for $n \equiv 5$

$$\begin{aligned} \mathcal{E}_5 = \frac{\sin(\xi h)}{h} [10 \sin^2 \xi + 6 \cos^2(\xi - \xi h) - 4 \cos^2(2\xi - \xi h)] \\ - 4 \sin(\xi h) \sin^2 \xi - 2 \cos(\xi h) \sin \xi \cos(3\xi - 2\xi h). \end{aligned}$$

Both functions are real analytic on $\mathbb{R} \times \mathbb{R}$ (with $\sin(\xi h)/h = \xi \operatorname{sinc}(\xi h)$) and odd in ξ .

Proof. For $n \equiv 4$ we have $\eta = 1$ and m odd, so $\psi \equiv \frac{\pi}{2} - \xi \pmod{\pi}$ and $\delta = \frac{\pi}{2} - \xi h$; for $n \equiv 5$ we have $\eta = -1$ and m even, so $\psi \equiv -\xi \pmod{\pi}$ and $\delta = -\frac{\pi}{2} + \xi h$. Substitute into F_0, F_1 (which are π -periodic in ψ) and use $s = \eta h$. $\square$

Theorem 4.10. *Let $n \equiv 4, 5 \pmod{6}$. On each open half of the central circle $E \neq 0$; more precisely $E > 0$ on the left half and $E < 0$ on the right half. Consequently φ is strictly monotone along each closed half, from one crossing to the other.*

Proof. On the open left half $A = \sin(m'w) > 0$, $B = \sin w > 0$, $C > 0$, and $\sin a \sin b \neq 0$. Indeed, for $n \equiv 4$ the zeros of $\sin(mv)$ nearest to $\pi/2$ are at distance $\pi/(2m)$, while $\sin(kv)$ vanishes at $\pi/2$ and next at distance $\pi/(m+1)$; for $n \equiv 5$, $\sin(mv)$ vanishes at $\pi/2$ and next at distance π/m , while the zeros of $\sin(kv)$ are at distance at least $\pi/(2m-2)$. All these distances exceed the half-width $2\pi/N = \pi/m'$ of J_0 . By Lemma A.2, $\mathcal{E}_n(\xi, h)/\xi^3 > 0$ on the whole parameter domain, so $E > 0$ on the open left half, and Lemma 4.2 gives $d\varphi/d\psi \neq 0$ away from the fold. At the fold Lemma 4.3 applies. On the right half $v = \frac{\pi}{2} + w$ corresponds to $-\xi$, and $E < 0$ because $\mathcal{E}_n$ is odd in ξ ; alternatively, use Lemma 4.7. A function on a closed arc whose derivative does not vanish in the interior is strictly monotone on the closed arc. $\square$

Proposition 4.11. *On each open half of the central circle, $\gamma_* < \gamma < \pi - \gamma_*$.*

Proof. On the left half, $\cos^2 \gamma = x^2 \sin^2 v / \sin^2(mv)$ by (4.1). For $n \equiv 4$ we have $\sin^2 v = \cos^2(\xi h)$, $\sin^2(mv) = \cos^2 \xi$ and $Q = -\sin(3\xi + \xi h) / \sin(\xi + \xi h)$; for $n \equiv 5$, $\sin^2(mv) = \sin^2 \xi$ and $Q = \cos(3\xi - \xi h) / \cos(\xi - \xi h)$. Since $\cos^2 \gamma_* = (2 + h) / (2 + 2h)$, resp. $1 - h/2$, clearing the (positive) denominators gives

$$\cos^2 \gamma_* - \cos^2 \gamma = \frac{\mathcal{N}_4(\xi, h)}{8(1 + h) \sin(\xi + \xi h) \cos^2 \xi}, \quad \text{resp.} \quad \cos^2 \gamma_* - \cos^2 \gamma = \frac{\mathcal{N}_5(\xi, h)}{4 \cos(\xi - \xi h) \sin^2 \xi},$$

with

$$\mathcal{N}_4 = 4(2 + h) \sin(\xi + \xi h) \cos^2 \xi - (2 + 2h)(\sin(\xi + \xi h) + \sin(3\xi + \xi h)) \cos^2(\xi h),$$

$$\mathcal{N}_5 = 2(2 - h) \cos(\xi - \xi h) \sin^2 \xi - (\cos(\xi - \xi h) - \cos(3\xi - \xi h)) \cos^2(\xi h).$$

The denominators are positive on the parameter domain, and Lemma A.3 shows $\mathcal{N}_n > 0$ for $\xi > 0$. Hence $\cos^2 \gamma < \cos^2 \gamma_*$ on the open left half, and the right half follows by Lemma 4.7. $\square$

The next proposition records how far φ moves along a half. Its proof needs the image of B (Proposition 3.9): $\varphi \equiv -3\gamma \pmod{2\pi}$ on the image of B for $n \equiv 4$, and $\varphi \equiv -\gamma$ for $n \equiv 5$. It also needs which crossing is the end of C_0^L on the branch $x > 0$: by Propositions 3.10 and 3.9 this is c_+ if $m \equiv 1 \pmod{4}$ ($n \equiv 4$) or $m \equiv 0 \pmod{4}$ ($n \equiv 5$), and c_- if $m \equiv 3$, resp. $m \equiv 2 \pmod{4}$.

Proposition 4.12. *Let $n \equiv 4, 5 \pmod{6}$. Along each half of the central circle, traversed from c_+ to c_- , a continuous lift of φ increases by*

$$\pi + 6\gamma_* \quad (n \equiv 4), \quad \text{resp.} \quad 3\pi + 2\gamma_* \quad (n \equiv 5).$$

The points of a half with $\varphi \in \pi\mathbb{Z}$ are two, resp. four, in number; exactly one, resp. two, of them lie on Δ , and these intersections are transverse.

Proof. As in the proof of Proposition 4.6, the points of the open left half with $\varphi \in \pi\mathbb{Z}$ are the points over the zeros $v_j = j\pi/n$ of $\sin nv$ in $(v_L, \pi/2)$, one on each branch, labelled 0 or π by the same parity rule.

The zeros. For $n \equiv 4$, $N = 4m + 2$ and $(2m - 1)n/N = \frac{3m}{2} - 1 + \frac{1}{4m+2}$; since m is odd, the only integer j with $(2m - 1)n/N < j < n/2$ is $j = \frac{3m-1}{2}$. For $n \equiv 5$, $N = 4m - 2$ and $(2m - 3)n/N = \frac{3m}{2} - 2 - \frac{1}{4m-2}$; since m is even, the integers in $((2m - 3)n/N, n/2)$ are $j = \frac{3m}{2} - 2$ and $j = \frac{3m}{2} - 1$.

The walk. Start at the end of C_0^L on the branch $x > 0$ and follow the half. By Theorem 4.10 the lift of φ is strictly monotone, so the points with $\varphi \in \pi\mathbb{Z}$ are met at consecutive multiples of π , and the label of the first one determines the direction of travel. The table lists the four cases; φ_0 is the starting value, and the zeros are met in the order of decreasing v on the branch $x > 0$ and increasing v on the branch $x < 0$.

case	start	φ_0	labels met	end value
$n \equiv 4, m \equiv 1 \pmod{4}$	c_+	$-3\gamma_*$	$0, \pi$	$\pi + 3\gamma_*$
$n \equiv 4, m \equiv 3 \pmod{4}$	c_-	$\pi + 3\gamma_*$	$\pi, 0$	$-3\gamma_*$
$n \equiv 5, m \equiv 0 \pmod{4}$	c_+	$-\gamma_*$	$0, \pi, 0, \pi$	$3\pi + \gamma_*$
$n \equiv 5, m \equiv 2 \pmod{4}$	c_-	$\pi + \gamma_*$	$\pi, 0, \pi, 0$	$-2\pi - \gamma_*$

For instance, in the first row $j = \frac{3m-1}{2}$ is odd, so the zero on the branch $x > 0$ has $\varphi \equiv 0$; from $\varphi_0 = -3\gamma_* \in [-\pi/2, 0)$ the first multiple of π reached is 0 only if φ increases. The zero on the branch $x < 0$ then has $\varphi = \pi$, and the end c_- has $\varphi \equiv -3(\pi - \gamma_*) \equiv \pi + 3\gamma_*$, which is first

reached at $\pi + 3\gamma_* < 2\pi$. The other rows are identical in structure; in each, the last step is at most $3\gamma_* \leq \pi/2$, so no further multiple of π is crossed. In every row the change from the c_+ end to the c_- end is $\pi + 6\gamma_*$, resp. $3\pi + 2\gamma_*$.

The right half is $\varsigma(C_0^L)$. If n is even, ς acts trivially on the pillowcase and the statement is the same. If n is odd, ς maps φ to $-\varphi - \pi$ and exchanges the ends with $\gamma = \gamma_*$ and $\gamma = \pi - \gamma_*$ (Lemma 4.7), so the change from the c_+ end is again $3\pi + 2\gamma_*$. The count of points on Δ is read off from the labels, and transversality follows from Theorem 4.10. $\square$

Finally we need the shape of the image of a half near a crossing. When n is even, ς fixes the crossings, reverses the direction of C_0 at them and acts trivially on the pillowcase, so the image of C_0 has zero velocity at $c_{\pm}$; the next lemma shows that this happens for all $n \equiv 4, 5$, and gives the leading term.

Lemma 4.13. *Let $n \equiv 4, 5 \pmod{6}$, and let $w = |v - \pi/2|$ along a half near a crossing $c \in \{c_+, c_-\}$. Then v is a regular parameter of C_0 at c , and on a lift*

$$(\gamma, \varphi) = (\gamma(c), \varphi(c)) \pm (V_\gamma, V_\varphi) w^2 + O(w^3),$$

with the sign $+$ at c_+ and $-$ at c_- , where

$$V_\gamma = \frac{\kappa h m^2}{\sin 2\gamma_*}, \quad V_\varphi = \frac{c_3 m^3}{8\sqrt{m'} k} \quad (n \equiv 4), \quad V_\varphi = \frac{c_3 m^2}{8\sqrt{m'}} \quad (n \equiv 5),$$

$\kappa = \frac{(h+2)(3h+1)}{6(h+1)}$, $c_3 = 4(h+1)(h+2)(h+3)$ for $n \equiv 4$, and $\kappa = \frac{(2-h)(2h+1)}{6}$, $c_3 = \frac{4}{3}(2-h)(11-2h)$ for $n \equiv 5$. In particular $V = (V_\gamma, V_\gamma + V_\varphi)$, the leading coefficient of (γ, θ) , is nonzero and has slope $1 + V_\varphi/V_\gamma > 1$; it is not parallel to the image of B , whose slope is -2 , resp. 0 .

Proof. The parameter v is regular at c because the tangent of C_0 at c has a nonzero v -component (Proposition 3.10). The expansion of $\cos^2 \gamma_* - \cos^2 \gamma$ in the proof of Proposition 4.11 begins with $\kappa h \xi^2$ (Lemma A.3), and $d \cos^2 \gamma / d\gamma = -\sin 2\gamma$; this gives V_γ , with γ moving into $(\gamma_*, \pi - \gamma_*)$ at both ends. For V_φ , insert the leading terms into (4.4): for $n \equiv 4$, $A \sim m'w$, $B \sim w$, $C \rightarrow 1$, $\sin^2 a \rightarrow 1$, $\sin^2 b \sim k^2 w^2$ and $E \sim c_3 m^2 w^3$ (Lemma A.2), so $|d\varphi/dw| = m|d\varphi/d\psi| \sim c_3 m^3 w / (4\sqrt{m'} k)$. For $n \equiv 5$, $\sin a \sim mw$ and $\sin^2 b \rightarrow 1$ instead, giving $|d\varphi/dw| \sim c_3 m^2 w / (4\sqrt{m'})$. The signs follow from Proposition 4.12: φ increases away from c_+ and decreases towards c_- . Since $V_\gamma > 0$ and $V_\varphi > 0$, the slope of V exceeds 1. $\square$

For example, for $n = 4$ one has $V_\gamma = 2/\sqrt{3}$ and $V_\varphi = 2\sqrt{3}$, so V has slope 4; the values of the leading coefficient of φ obtained from the formula agree with direct evaluation of the restriction map for $n = 4, 5, 10, 11, 16, 17, 41$ to at least four significant digits.

5. SMALL PERTURBATIONS

In this section we describe the perturbed variety W_ϵ , $\epsilon = (\epsilon_A, \epsilon_B)$, for small ϵ , together with the restriction of Π to it. Hedden, Herald and Kirk show that for Sard-generic small ϵ the space $R_\pi(Y, T)$ is a compact 1-manifold with two boundary points, and that the restriction map is a restricted immersed 1-manifold except possibly for fishtails [HHK18, Thm. 10.1]. Near the two abelian representations their argument uses [HHK18, Thm. 8.4], which is proved by combining the description of the unperturbed variety near $r_{\pm}$ [HHK18, Prop. 8.1] with the persistence of the abelian points under perturbation [HHK18, Prop. 8.3]. The analysis of Section 6 needs more precise information: a description of W_ϵ near the abelian points that is uniform in ϵ , the local structure of W_ϵ near the two double points of W when $n \equiv 4, 5 \pmod{6}$, and C^1 control elsewhere. We treat the three regions in turn and then assemble them (Proposition 5.7).

Throughout, $\Psi = \Psi(\epsilon; u, v, \tau) = (\Psi_1, \Psi_2)$ denotes the pair of functions of [HHK18, (33)–(34)],

$$\begin{aligned}\Psi_1 &= \cos \Theta_1 \cos \Phi_1 - \tau \sin \Theta_1 \sin \Phi_1, & \Psi_2 &= \cos \Theta_2 \cos \Phi_2 - \tau \sin \Theta_2 \sin \Phi_2, \\ \Phi_1 &= (s+p)u + (q-r)\epsilon_A \sin u, & \Phi_2 &= su - r\epsilon_A \sin u, \\ \Theta_1 &= (q-r)v - (s+p)\epsilon_B \sin v, & \Theta_2 &= -rv - s\epsilon_B \sin v,\end{aligned}$$

so that $W_\epsilon = \Psi(\epsilon; \cdot)^{-1}(0) \subset [0, \pi]^2 \times [-1, 1]$. At $\epsilon = 0$, Ψ is $(E1, E2)$ for $\eta = 1$ and $(E2, E1)$ for $\eta = -1$ (Lemma 3.1). The map $\Pi = \Pi_\epsilon$ sends a point of W_ϵ to the restriction of the corresponding representation to the boundary sphere; it is computed from the elements a, b, c of [HHK18, (31)] and depends smoothly on (ϵ, u, v, τ) where it is defined. We write $\tilde{\Pi}$ for a lift to $\mathbb{R}^2$.

5.1. The arc near its ends. By Proposition 3.6, the ends of the arc B are the two abelian points of $R(Y, T)$. For n odd (m even) they are the ends $(\frac{\pi}{2}, \frac{\pi}{2}, \pm 1)$ of the fiber B . For n even (m odd) they are the segments $\{u = 0, v = \frac{\pi}{2}\} \times [-1, 1]$ and $\{u = \pi, v = \frac{\pi}{2}\} \times [-1, 1]$, each of which is a single point of $R(Y, T)$ [HHK18, Thm. 10.2]. We introduce coordinates in which these points are regular.

Write $\rho(A_1) = e^{uQ_A}$ and $\rho(B_1) = e^{vQ_B}$ with Q_A, Q_B traceless unit quaternions and $\tau = Q_A \cdot Q_B$, as in [HHK18, Thm. 10.2]. The representation is abelian exactly when the two one-parameter subgroups commute.

- *n odd.* Conjugate so that $Q_A = \mathbf{i}$, and write $Q_B = \tau \mathbf{i} + y \mathbf{j}$ with $y \in \mathbb{R}$ and $\tau = \pm \sqrt{1 - y^2}$. Conjugation by $\mathbf{i}$ changes the sign of y and fixes u, v . The coordinates of the chart are (u, v, w) with $w = y^2 = 1 - \tau^2$.
- *n even.* Conjugate so that $Q_B = \mathbf{j}$, and put $\xi = uQ_A \in \mathbb{R}^3$. The exponents Φ_1, Φ_2 and $pu + q\epsilon_A \sin u$ are odd functions of u , so every factor $e^{\Phi Q_A} = \cos \Phi(|\xi|) + \frac{\sin \Phi(|\xi|)}{|\xi|} \xi$ entering [HHK18, (31)] is a real-analytic function of ξ . Rotating about $\mathbf{j}$ we may take $\xi = (y, x, 0)$ with $y \in \mathbb{R}$; then $x = u\tau$, $y^2 = u^2(1 - \tau^2)$, and rotation by π about $\mathbf{j}$ changes the sign of y . The coordinates of the chart are (x, w, v) with $w = y^2$. Near $u = \pi$ the same construction applies to $\xi = (u - \pi)Q_A$; the exponents become $k\pi$ plus an odd function, which contributes a sign $(-1)^k$.

In both cases the abelian locus is $\{w = 0\}$, the functions Ψ are real-analytic and even in y , hence real-analytic functions of w (and of ϵ), and the boundary representation (a, b, c) is a real-analytic function of the signed coordinate y . For n even, $(x, w) \mapsto (u, \tau)$ is a diffeomorphism from $\{w > 0\}$ onto $\{u > 0, |\tau| < 1\}$ with inverse $u = \sqrt{x^2 + w}$; it collapses the segment $\{u = 0\}$ to the point $(x, w) = (0, 0)$, as in [HHK18, Thm. 10.2].

Lemma 5.1 (Corner charts). *Let $r_\pm(0)$ be one of the two abelian points of $R(Y, T)$. There are a neighbourhood U of $r_\pm(0)$, a number $y_0 > 0$ and a neighbourhood $\mathcal{O}$ of $0 \in \mathbb{R}^2$ such that for every $\epsilon \in \mathcal{O}$:*

- (1) $R_\pi(Y, T) \cap U$ is a single embedded half-open arc $\{\Gamma_\epsilon(y) : 0 \leq y < y_0\}$, with $\Gamma_\epsilon(0) = r_\pm(\epsilon)$ the perturbed abelian point, and Γ is real-analytic in (y, ϵ) ;
- (2) Π maps $r_\pm(\epsilon)$ to the corner $\Pi(r_\pm(0))$, and with a suitable lift,

$$\tilde{\Pi}(\Gamma_\epsilon(y)) = y c(\epsilon) + y^3 R(y, \epsilon),$$

where $c(\epsilon) \in \mathbb{R}^2$ and R are real-analytic and $c(0)$ is a nonzero vector along the image of B , of slope $2, 4, -2, 0$ for $n \equiv 1, 2, 4, 5 \pmod{6}$ (Proposition 3.9).

Proof. On $\{w = 0\}$ the equations $\Psi = 0$ reduce to the traceless conditions for abelian representations. For n odd these read $\cos(\Phi_1 \pm \Theta_1) = \cos(\Theta_2 \pm \Phi_2) = 0$, with the sign of τ ; their Jacobian with respect to (u, v) is

$$\pm(\Phi'_1 \Theta'_2 - \Theta'_1 \Phi'_2) + O(\epsilon) = \pm((s+p)(-r) - (q-r)s) + O(\epsilon) = \mp(pr + qs) + O(\epsilon) = \mp 1 + O(\epsilon),$$

since $\sin(\Phi_1 \pm \Theta_1)$ and $\sin(\Theta_2 \pm \Phi_2)$ are ± 1 at the abelian point. For n even, at $(x, w) = (0, 0)$ the derivatives of Ψ_k with respect to x and v are $-\Phi'_k(0) \sin \Theta_k$ and $-\Theta'_k \sin \Theta_k$, with $\sin \Theta_k = \pm 1$ at $v = \frac{\pi}{2}$, so the Jacobian with respect to (x, v) is again $\pm(\Phi'_1 \Theta'_2 - \Theta'_1 \Phi'_2) + O(\epsilon) = \mp 1 + O(\epsilon)$. (At the chart centred at $u = \pi$ the core term $\Phi'_1 \Theta'_2 - \Theta'_1 \Phi'_2$ changes sign; only its absolute value matters.)

By the implicit function theorem, the two coordinates other than w are real-analytic functions of (w, ϵ) near $(0, 0)$. Hence W_ϵ near the abelian point is the graph of a real-analytic map on $\{w \geq 0\}$, and its intersection with $\{w = 0\}$ is the single perturbed abelian point $r_\pm(\epsilon)$. Passing to $y = \sqrt{w}$ and to $R_\pi(Y, T)$ gives (1). (In box coordinates there is one exception: for n even, if the perturbed abelian point has $x = 0$, for instance when $\epsilon_B = 0$, then $W_\epsilon \cap U$ also contains the collapsed segment $\{u = 0\} \times [-1, 1]$, which is the single point $r_\pm(\epsilon)$ of $R_\pi(Y, T)$.)

For (2), the boundary representation is real-analytic in (y, ϵ) and is a corner representation at $y = 0$: the traceless elements a, b, c are all equal to $\pm$ one unit vector. Write $\pm b - a = y b_1 + O(y^2)$, with the sign $+$ at the corner $(0, 0)$ and $-$ at $(\pi, 0)$, where $b = -a$, and $c - a = y c_1 + O(y^2)$. Since a, b, c lie on a common great circle, b_1 and c_1 are parallel to a unit vector e orthogonal to a ; in the normalized frame $a = \mathbf{i}$, $b = e^{\gamma \mathbf{k}} \mathbf{i}$, $c = e^{\theta \mathbf{k}} \mathbf{i}$ this gives the lift $(\tilde{\gamma}, \tilde{\theta}) = y(b_1 \cdot e, c_1 \cdot e) + O(y^2)$. Changing the sign of y is a conjugation, so it fixes the point of P ; on the branched double cover $\mathbb{R}^2 \rightarrow P$ near the lattice point it acts as the deck involution -1 . With the lift chosen accordingly, $\tilde{\Pi}(\Gamma_\epsilon(y))$ is an odd real-analytic function of y , which gives the form $y c(\epsilon) + y^3 R$. At $\epsilon = 0$ the half-arc is the end of B , whose image is the straight segment of Proposition 3.9, traversed with $\gamma = |y| + O(y^3)$ (for n odd, $\gamma = \arccos \tau$; for n even, $\gamma = u$ along $\tau = 0$). So $c(0) \neq 0$ points along the image of B , and $b_1(\epsilon) \neq 0$ for small ϵ , which makes e well defined. $\square$

Remark 5.2. The perturbed abelian points have a simple implicit description. For an abelian representation put $\rho(A_1) = e^{\mu_1 \mathbf{i}}$ and $\rho(B_1) = e^{\mu_2 \mathbf{i}}$ in a common maximal torus. By [HHK18, (32)], $\rho(A_2) = e^{\epsilon_A \sin \mu_1 \mathbf{i}}$ and $\rho(B_2) = e^{\epsilon_B \sin \mu_2 \mathbf{i}}$, and [HHK18, (31)] converts the traceless conditions into $J\mu + E \sin \mu \equiv \text{const}$, with $J = \begin{pmatrix} s+p & q-r \\ s & -r \end{pmatrix}$ of determinant $-(pr + qs) = -1$ and $J^{-1}E = \begin{pmatrix} 0 & -\epsilon_B \\ \epsilon_A & 0 \end{pmatrix}$. Hence

$$\mu_1 = \mu_1^0 + \epsilon_B \sin \mu_2, \quad \mu_2 = \mu_2^0 - \epsilon_A \sin \mu_1,$$

where μ^0 is the unperturbed abelian point. For n even and the end over $u = 0$ this gives $u = |\epsilon_B| + O(|\epsilon|^3)$ and $\tau = \text{sign } \epsilon_B$: the perturbed arc ends on a face $\tau = \pm 1$ of the box rather than on the collapsed segment, unless $\epsilon_B = 0$.

Corollary 5.3. *For $\epsilon \in \mathcal{O}$ the following hold near the corner $(0, 0)$, with constants independent of ϵ .*

- (1) *The limiting slope of L_1 at r_+ is the slope of the image of B plus $O(|\epsilon|)$; in particular it is bounded away from 1.*
- (2) *On the lift through the lattice point, $\varphi = y(c_2(\epsilon) - c_1(\epsilon)) + O(y^3)$ with $c_2 - c_1$ bounded away from 0. Hence φ is strictly monotone on $0 < y < y_0$, negative for $n \equiv 4, 5$ and positive for $n \equiv 1, 2 \pmod{6}$: the arc leaves the corner below Δ in the first case and above Δ in the second.*
- (3) *The full preimage of $L_1(R_\pi(Y, T) \cap U)$ near the lattice point is a single embedded real-analytic curve through it. In particular the arc has no self-intersections in $\Pi(U)$.*
- (4) *For every sufficiently small earring parameter (the parameter of (2.1), not the pair (ϵ_A, ϵ_B)), L_0 meets $L_1(R_\pi(Y, T) \cap U)$ in exactly one point, the generator r_+ of [HHK18, Lemma 5.1]. It lies on the strand S_- for $n \equiv 4, 5$ and on S_+ for $n \equiv 1, 2$.*

The same statements, without (4), hold at the other end of the arc, whose image is the corner $(\pi, 0)$.

Proof. Items (1)–(3) follow from Lemma 5.1(2): the derivative of the lift is $c(\epsilon) + O(y^2)$, and $c_2(0) - c_1(0)$ is the slope of B 's image minus one, which is $1, 3, -3, -1$ in the four classes. For (4), the strands of L_0 pass the corner on either side of Δ , at distance $\sqrt{2}$ times the earring parameter (Section 2), so a curve leaving the corner with slope bounded away from 1 and with φ monotone meets exactly one of them, once, near the corner: the strand S_- below Δ or the strand S_+ above it. The corner $(\pi, 0)$ is not an endpoint of Δ . $\square$

5.2. The double points. Let $n \equiv 4, 5 \pmod{6}$ and let c be one of the two points where the central circle C_0 meets B . By Proposition 3.11, $d_{(u,v,\tau)}\Psi(c)$ has rank one, its kernel is spanned by the tangent lines T_B and T_C of B and C_0 at c , and the restriction to this kernel of $\ell_c \cdot D^2\Psi(c)$, where ℓ_c is a nonzero covector vanishing on the image of $d\Psi(c)$, is a nondegenerate indefinite quadratic form. Choose coordinates (s, t, ϱ) centred at c with $\partial_s = T_B$ and $\partial_t = T_C$ at c . The component of Ψ transverse to $\ker \ell_c$ has nonzero ϱ -derivative at c , so by the implicit function theorem it can be solved for $\varrho = \varrho(s, t, \epsilon)$. Near c , W_ϵ is then the zero set of the reduced function

$$g(s, t; \epsilon) = \ell_c \cdot \Psi(\epsilon; s, t, \varrho(s, t, \epsilon)),$$

whose Hessian at $(0, 0; 0)$ is the quadratic form above. By the implicit function theorem applied to dg , $g(\cdot; \epsilon)$ has a unique critical point near 0 for small ϵ , depending analytically on ϵ , with critical value $\kappa_c(\epsilon)$. The Morse lemma, whose proof via Hadamard's lemma (see [Mil63, Lemma 2.2]) goes through with analytic dependence on the parameter ϵ , gives coordinates (s', t') , depending analytically on ϵ , with

$$g = \kappa_c(\epsilon) + s't'.$$

At $\epsilon = 0$ we choose them so that $\{t' = 0\}$ is B and $\{s' = 0\}$ is C_0 near c . We have $\kappa_c(0) = 0$ and $d\kappa_c(0) = \ell_c \cdot \partial_\epsilon \Psi(0; c)$.

Lemma 5.4. *For each double point c , $\kappa_c(\epsilon) = \lambda_c \epsilon_A + \mu_c \epsilon_B + O(|\epsilon|^2)$ with $\mu_c \neq 0$. For $n \equiv 4 \pmod{6}$, $\lambda_c = 0$, and $\kappa_c(\epsilon_A, 0) = 0$ for all ϵ_A .*

Proof. The function Ψ_k depends on v and on ϵ_B only through Θ_k . At $v = \frac{\pi}{2}$ and $\epsilon = 0$, write $X_k = \partial \Psi_k / \partial \Theta_k(c)$. Then

$$\partial_v \Psi(c) = ((q - r)X_1, -rX_2), \quad \partial_{\epsilon_B} \Psi(c) = (-(s + p)X_1, -sX_2),$$

and the determinant of these two vectors is $-X_1 X_2 ((q - r)s + r(s + p)) = -X_1 X_2 (pr + qs) = -X_1 X_2$. Both components of $\partial_v \Psi(c)$ are nonzero. For $n \equiv 4$, $\partial_v(E1, E2)(c) = (-m\omega_1 \cos u_1, -m'\omega_2 \cos 2u_1)$ with $\cos 2u_1 = \frac{m}{m+1}$. For $n \equiv 5$, $\partial_v(E1, E2)(c) = (-\tau m\omega_1, m'\omega_2)$ with $\tau = \pm \sqrt{1 - 1/(2m)}$. Here $\omega_1, \omega_2 = \pm 1$ are as in Proposition 3.11. Hence $X_1 X_2 \neq 0$. Since the image of $d\Psi(c)$ is spanned by $\partial_v \Psi(c)$ and $\partial_{\epsilon_B} \Psi(c)$ is not proportional to it, $\mu_c = \ell_c \cdot \partial_{\epsilon_B} \Psi(c) \neq 0$.

For $n \equiv 4$ the double points are $(u_1, \frac{\pi}{2}, 0)$ and $(\pi - u_1, \frac{\pi}{2}, 0)$, where $\partial_u \Psi = 0$. Since Ψ_k depends on u and ϵ_A only through Φ_k , and $\partial_u \Phi_k = s + p$ resp. s is nonzero at $\epsilon = 0$, this gives $\partial \Psi_k / \partial \Phi_k(c) = 0$ and hence $\partial_{\epsilon_A} \Psi(c) = 0$, so $\lambda_c = 0$. Finally, for $n \equiv 4$ and $\epsilon_B = 0$ the line $\{v = \frac{\pi}{2}, \tau = 0\}$ lies in W_ϵ for every ϵ_A , because $\cos \Theta_1 = \cos((q - r)\frac{\pi}{2}) = 0$ and $\cos \Theta_2 = \cos(r\frac{\pi}{2}) = 0$ with $q - r = m$ and $r = m'$ odd. By Remark 3.12, $d\Psi$ has rank at most one at some point $x \in B$ near c (in general $x \neq c$). On the zero set of the component of Ψ solved for ϱ , the rank of $d\Psi$ is one plus the rank of dg , so $dg(x) = 0$, and $g(x) = 0$ because $x \in W_{(\epsilon_A, 0)}$. Hence x is the unique critical point of $g(\cdot; (\epsilon_A, 0))$ near c , and $\kappa_c(\epsilon_A, 0) = g(x) = 0$. $\square$

Numerically, $\mu_c = \pm \frac{1}{4}$ for $n = 4, 5$ and $|\mu_c| = 0.1179, 0.0772, 0.0574$ for $n = 10, 16, 22$ and for $n = 11, 17, 23$, with ℓ_c normalized to unit length; for $n \equiv 5$ also $|\lambda_c| = \sqrt{3}/4 \approx 0.433$ for $n = 5$ and $|\lambda_c| \approx 0.441, 0.443, 0.444$ for $n = 11, 17, 23$. For $T(3, 4)$, Lemma 5.4 and Remark 3.12 show that the perturbation along ϵ_A alone does not resolve the double points. This differs from [HHK18, §11.6], where $R_{\epsilon_A, 0}(Y, T)$ is treated as a restricted immersed 1-manifold, the union of an arc and a circle, for every small nonzero ϵ_A ; by Lemma 5.4 it is singular at the double points.

Definition 5.5. For $a > 0$ let $\mathcal{C}_a$ be the set of $\epsilon \in \mathbb{R}^2$ with $|\lambda_c \epsilon_A + \mu_c \epsilon_B| \geq a|\epsilon|$ at both double points. For $n \equiv 1, 2 \pmod{6}$ put $\mathcal{C}_a = \mathbb{R}^2$. By Lemma 5.4, for a small $\mathcal{C}_a$ is a closed cone containing the ϵ_B -axis; for $n \equiv 4$ it is $\{|\epsilon_B| \geq a'|\epsilon|\}$ for some $a' > 0$.

For $\epsilon \in \mathcal{C}_a$ small we have $|\kappa_c(\epsilon)| \geq \frac{a}{2}|\epsilon|$, so the zero set $\{s't' = -\kappa_c(\epsilon)\}$ is smooth. It consists of two arcs, each joining a half-branch of B at c to a half-branch of C_0 at c . Which pair is joined depends on the sign of $\kappa_c(\epsilon)$.

Lemma 5.6. *Let $a > 0$. There are $r_0 > 0$ and $\epsilon_0 > 0$ with the following property. For $\epsilon \in \mathcal{C}_a$ with $0 < |\epsilon| < \epsilon_0$, let U_c be the set of points of a fixed neighbourhood of c whose coordinates (s, t) have Morse coordinates (s', t') with $|s'| \leq r_0$ and $|t'| \leq r_0$; it depends on ϵ , but it contains and is contained in fixed neighbourhoods of c . Then each of the two arcs of $W_\epsilon \cap U_c$ is mapped by Π to an embedded arc with nowhere vanishing velocity. Along such an arc φ has at most one critical point. It has exactly one if and only if the two half-branches it joins move φ in the same sense away from c , and in that case the critical value differs from $\varphi(c)$ by $o(1)$ as $\epsilon \rightarrow 0$.*

Proof. By Proposition 3.9, $\Pi(s', 0) = \Pi(c) + s'P_B + O(s'^2)$ at $\epsilon = 0$ with $P_B \neq 0$ along the image of B . By Lemma 4.13, $\Pi(0, t') = \Pi(c) + t'^2V + O(t'^3)$ with $V \neq 0$, $V \nparallel P_B$, and $d\varphi(V) \neq 0$; also $d\varphi(P_B) \neq 0$, since the slope of B 's image is not 1. Extend Π_ϵ smoothly to a neighbourhood of c . Then, in the coordinates (s', t') ,

$$\Pi_\epsilon(s', t') = \Pi(c) + s'P_B + t'^2V + O(s'^2 + |s't'| + |t'|^3 + |\epsilon|(|s'| + |t'| + 1)),$$

with the corresponding bounds on first derivatives. Parametrize an arc of $W_\epsilon \cap U_c$ by t' , with $s' = -\kappa/t'$, $\kappa = \kappa_c(\epsilon)$, and $|\kappa|/r_0 \leq |t'| \leq r_0$. Its velocity is

$$\frac{d}{dt'}\Pi_\epsilon = \frac{\kappa}{t'^2}P_B + 2t'V + e(t'), \quad |e(t')| \leq C\left(\frac{|\kappa|}{t'^2}(r_0 + |\epsilon|) + \frac{|\kappa|}{|t'|} + t'^2 + |\epsilon|\right).$$

Put $A(t') = |\kappa|/t'^2 + |t'|$. Since P_B and V are linearly independent, the main term has length at least $c_0A(t')$. Each error term is at most a small multiple of A : $|\kappa|/|t'| \leq r_0A$, $t'^2 \leq r_0A$, and $|\epsilon| \leq |\epsilon|A/(c'|\kappa|^{1/3}) \leq C'|\epsilon|^{2/3}a^{-1/3}A$, because $A \geq c'|\kappa|^{1/3}$ and $|\kappa| \geq \frac{a}{2}|\epsilon|$. Hence, using also $|\epsilon| \leq |\epsilon|^{2/3}a^{-1/3}$ for $|\epsilon| \leq 1/a$,

$$|e(t')| \leq C''(r_0 + |\epsilon|^{2/3}a^{-1/3})A(t'),$$

with C'' independent of ϵ , a and r_0 . Along the arc the signs of κ and t' are fixed, so the main term lies in the closed convex cone K spanned by $\text{sign}(\kappa)P_B$ and $\text{sign}(t')V$; its opening θ_K is $\angle(P_B, V)$ or $\pi - \angle(P_B, V)$, in either case less than π . Fix $\delta \in (0, 1)$ with $2 \arcsin \delta < \pi - \theta_K$ for all four choices of the signs. For r_0 and $|\epsilon|$ small, $|e| \leq \delta c_0A$, so the velocity never vanishes and makes an angle at most $\arcsin \delta$ with K : it lies in a fixed open sector of angle $\theta_K + 2 \arcsin \delta < \pi$. A curve whose velocity stays in such a sector is a graph over a line, hence embedded.

For the critical points, apply $d\varphi$: with $p = d\varphi(P_B)$ and $q = d\varphi(V)$ the derivative of φ along the arc is $\kappa p/t'^2 + 2qt'$ plus an error bounded as above. On the arc $\text{sign}(s')\text{sign}(t') = -\text{sign} \kappa$, and the main term vanishes exactly once, transversally, at $t'^3 = -\kappa p/(2q)$, if $\text{sign}(ps') = \text{sign}(q)$; that is, if φ increases away from c along both joined half-branches or decreases along both. Otherwise

the main term has constant sign and is at least a fixed multiple of A . The error does not change the number of zeros. To see this, put $y = 3 \log |t'| - \log |\kappa|$, so that $A = (|\kappa|/t'^2)(1 + e^y)$. The main term divided by A is $(\sigma_1 p + 2\sigma_2 q e^y)/(1 + e^y)$ with $\sigma_1 = \text{sign } \kappa$ and $\sigma_2 = \text{sign } t'$. It does not depend on κ , it has at most one zero, at which its y -derivative is bounded away from 0 in terms of p and q only, and it is bounded away from 0 outside a fixed interval of y . The error e_φ in the derivative of φ satisfies $|e_\varphi| \leq C''(r_0 + |\epsilon|^{2/3} a^{-1/3})A$ as above, and the same termwise estimates, applied to the second derivatives of Π_ϵ in the coordinates (s', t') , which are bounded uniformly in ϵ , give $|t' e'_\varphi(t')| \leq C''(r_0 + |\epsilon|^{2/3} a^{-1/3})A$ after enlarging C'' . Since $dt'/dy = t'/3$ and $|dA/dy| \leq A$, the error divided by A and its y -derivative are at most a fixed multiple of $r_0 + |\epsilon|^{2/3} a^{-1/3}$, which is small for r_0 and $|\epsilon|$ small. So the number of zeros is that of the main term. The critical point lies at $|t'| = O(|\kappa|^{1/3})$, so φ there is $\varphi(c) + o(1)$. $\square$

5.3. Global structure.

Proposition 5.7. *Let $a > 0$. For every $\epsilon \in \mathcal{C}_a$ with $0 < |\epsilon|$ sufficiently small:*

- (1) $R_\pi(Y, T)$ is a compact 1-manifold with two boundary points $r_\pm(\epsilon)$, and L_1 is an immersion into P^* away from $r_\pm(\epsilon)$.
- (2) Away from $U_{c_\pm}$ and from the corner charts of Lemma 5.1, W_ϵ is C^1 -close to W : it is the image of a C^1 -small normal graph over W , with Π_ϵ correspondingly C^1 -close to Π .
- (3) For $n \equiv 1, 2 \pmod{6}$ the components of $R_\pi(Y, T)$ are an arc B_ϵ close to B , from $r_+(\epsilon)$ to $r_-(\epsilon)$, and circles $C_{l,\epsilon}$ close to the non-central circles, one for each.
- (4) For $n \equiv 4, 5 \pmod{6}$ the same holds for the non-central circles, while $B \cup C_0$ is replaced by its resolution at the two double points. Label the pieces of B cut out by c_+, c_- as B_0 (from r_+), B_1 (between the double points) and B_2 (to r_-), and the halves of C_0 as C_0^L, C_0^R . Then either
 - the arc is $B_0 \cdot C_0^X \cdot B_2$ and there is one further circle $B_1 \cdot (C_0^Y)^{-1}$, or
 - the arc is $B_0 \cdot C_0^X \cdot B_1^{-1} \cdot C_0^Y \cdot B_2$ and there is no further circle,
 where $\{X, Y\} = \{L, R\}$ and each product denotes the resolved curve, which is C^1 -close to the listed pieces away from $c_\pm$ and is described by Lemma 5.6 near them.

Proof. Since Ψ depends continuously on ϵ and the box is compact, W_ϵ lies in any prescribed neighbourhood of W once $|\epsilon|$ is small. Cover W by three kinds of neighbourhoods:

- the corner charts U of Lemma 5.1;
- the neighbourhoods $U_{c_\pm}$ of Lemma 5.6 (only for $n \equiv 4, 5$);
- a neighbourhood of the remaining compact part K of W , on which $d_{(u,v,\tau)}\Psi$ has rank two (Lemma 3.8).

On a neighbourhood of K the implicit function theorem shows that W_ϵ is a smooth curve C^1 -close to W , uniformly in ϵ . On U and $U_{c_\pm}$ the structure is given by Lemmas 5.1 and 5.6. On overlaps the descriptions agree by the uniqueness part of the implicit function theorem. At $\epsilon = 0$ only the abelian points are mapped to corners: the interior of B is mapped to the interior of a straight segment between two corners (Proposition 3.9), and the circles and the two halves of C_0 avoid the edges $\gamma \in \pi\mathbb{Z}$ (Proposition 4.5). Hence for small ϵ every point of W_ϵ whose image is near a corner lies in a corner chart. This proves (2) and the first part of (1). The restriction map is an immersion wherever $R_\pi(Y, T)$ is a smooth 1-manifold [HHK18, p. 46, after Thm. 8.4]; in the three regions this is also visible directly from Lemmas 5.1 and 5.6 and from C^1 -closeness to $\Pi|_W$, which is an immersion on K .

The list of components in (3) and (4) is read off from the cover. Each non-central circle is a smooth component of W contained in K , so it persists. For $n \equiv 1, 2$, W has no double points (Proposition 3.10(1)), and B persists as an arc ending at the perturbed abelian points. For

$n \equiv 4, 5$, at each double point the resolution joins each half-branch of B to a half-branch of C_0 . If the half-branches B_0 and B_2 are joined to the same half of C_0 , the result is the first alternative; otherwise it is the second. $\square$

Remark 5.8. Both alternatives in Proposition 5.7(4) are compatible with the local analysis, and Section 6 shows that the conclusions of Theorem 1.1 do not depend on which occurs. Which one occurs is decided by the signs of $\kappa_{c_+}(\epsilon)$ and $\kappa_{c_-}(\epsilon)$. The zero set $\{s't' = -\kappa_c\}$ lies in the two quadrants with $\text{sign}(s't') = -\text{sign } \kappa_c$, so changing the sign of κ_c changes which half-branch of B is joined to which half-branch of C_0 at c ; changing one of the two signs therefore changes the alternative, and changing both does not. For $n \equiv 4 \pmod{6}$ and small $\epsilon \in \mathcal{C}_a$ the signs are those of $\mu_{c_\pm} \epsilon_B$ (Lemma 5.4 and Definition 5.5), and replacing ϵ_B by $-\epsilon_B$ changes both of them. Hence for $n \equiv 4$ the alternative is the same for all small $\epsilon \in \mathcal{C}_a$. We have not determined it by hand; it was the first alternative in all our computations. For $n \equiv 5 \pmod{6}$ our computations show both alternatives in $\mathcal{C}_a$. The first occurs near the ϵ_B -axis, for instance for $n = 5$ at $(\epsilon_A, \epsilon_B) = (0, 0.04)$ and for $n = 11, 17$ at $(0, 0.03)$, where the arc ends at $(\pi, 4\pi)$ and carries 5 generators. The second occurs in the sector containing the ϵ_A -axis, for instance at all the perturbations of Remark 6.12, which satisfy $|\lambda_c \epsilon_A| > 2|\mu_c \epsilon_B|$ at both double points. For $n = 5$ the two alternatives are separated exactly by the lines $|\epsilon_B| = \sqrt{3}|\epsilon_A|$: for small $\epsilon \in \mathcal{C}_a$ the first occurs when $|\epsilon_B| > \sqrt{3}|\epsilon_A|$ and the second when $|\epsilon_B| < \sqrt{3}|\epsilon_A|$ [Wue26b, Prop. 4.10], in agreement with the samples above. The cone condition $\epsilon \in \mathcal{C}_a$ is used only in Lemma 5.6, to keep $|\kappa_c(\epsilon)|$ large compared with $|\epsilon|^3$; it excludes, in particular, perturbations with $\epsilon_B = 0$ for $n \equiv 4 \pmod{6}$, for which the double points are not resolved at all (Lemma 5.4).

6. THE PILLOWCASE COMPLEX

We now assemble the results of Sections 3–5 into a description of the complex $C(L_0, L_1)$ and prove Theorems 1.1 and 1.3. Throughout, $n \geq 4$ is coprime to 3, the decomposition is the one fixed in Section 3, π denotes the holonomy perturbation with sufficiently small parameters (ϵ_A, ϵ_B) in the cone $\mathcal{C}_a$ of Definition 5.5, for a fixed $a > 0$ (we write π only as a subscript, in $R_\pi(Y, T)$ and $H^\natural(Y, T, \pi)$), and the earring parameter $\epsilon > 0$ is chosen sufficiently small after the perturbation has been fixed. All statements about intersection points with L_0 hold for every sufficiently small ϵ .

6.1. The curves and their lifts. For $n \equiv 4, 5 \pmod{6}$ let c_+ and c_- be the two points of $C_0 \cap B$, labelled so that $\Pi(c_+)$ and $\Pi(c_-)$ are the points of the image of B with $\gamma = \gamma_*$ and $\gamma = \pi - \gamma_*$ (Proposition 3.10). They cut B into three arcs: B_0 , joining the endpoint of B that maps to the corner $(0, 0)$ to c_+ ; B_1 , joining c_+ to c_- ; and B_2 , joining c_- to the endpoint that maps to $(\pi, 0)$. Each half of the central circle joins c_+ to c_- , and we orient both halves from c_+ to c_- . We use the following facts.

- (F1) The image of B is a straight segment from $(0, 0)$ to $(\pi, 0)$ in P , and along the lift of B starting at $(0, 0)$ into $\{\gamma > 0\}$ one has $\varphi = \gamma, 3\gamma, -3\gamma, -\gamma$ for $n \equiv 1, 2, 4, 5 \pmod{6}$ (Proposition 3.9).
- (F2) Along each half of C_0 , φ is strictly monotone with $d\varphi \neq 0$ at interior points; on the lift starting at the lift of c_+ , it changes by $\pi + 6\gamma_*$ ($n \equiv 4$) or $3\pi + 2\gamma_*$ ($n \equiv 5$); and $\gamma_* < \gamma < \pi - \gamma_*$ at interior points (Theorem 4.10 and Propositions 4.12, 4.11). Here $0 < \gamma_* \leq \pi/6$.
- (F3) Along each non-central circle φ is strictly monotone with $d\varphi \neq 0$ and changes by $\pm 4\pi$ over one period, and the circle misses the left and right edges (Theorem 4.4 and Propositions 4.5, 4.6).

(F4) For sufficiently small $(\epsilon_A, \epsilon_B) \in \mathcal{C}_a$ (Section 5): $R_\pi(Y, T)$ consists of circles C'_l , C^1 -close to the non-central circles, and of the curves obtained from $B \cup C_0$ by resolving the two double points c_+, c_- ; away from c_+, c_- and the abelian endpoints, the perturbed curves are C^1 -close to W (Proposition 5.7). Near c_+ and c_- the image of the resolved curve is an immersed rounded corner on which φ has at most one critical point; it has one exactly when the two joined half-branches move φ in the same sense away from c_+ or c_- (Lemma 5.6). Near the corners $(0, 0)$ and $(\pi, 0)$, the arc is described by the corner chart lemma (Lemma 5.1 and Corollary 5.3): its limiting slope differs from that of B by $O(|\epsilon_A| + |\epsilon_B|)$, it leaves $(0, 0)$ on the same side of Δ_0 as B , and $|d\varphi|$ is bounded below in arclength near the corner.

At each of c_+, c_- the perturbation selects one of the two resolutions of a transverse double point. The four possible combinations produce two combinatorial types. At c_+ the arc B_0 is joined to one half of C_0 and B_1 to the other; at c_- the same holds for B_2 and B_1 . If B_0 and B_2 are joined to the same half, the other half closes up with B_1 ; otherwise the arc component runs through B_1 . Write H for the half joined to B_0 and H' for the other. In the *joined* resolution $R_\pi(Y, T)$ contains a single arc built from all of $B \cup C_0$; in the *split* resolution it contains an arc and a further circle, the *remnant circle*, which contains B_1 . Explicitly, up to the C^1 -small changes of (F4),

$$\begin{aligned} n \equiv 4, \text{ split: } & A_0 = B_0 \cdot H \cdot B_2, & \text{remnant circle } B_1 \cdot H'^{-1}; \\ n \equiv 4, \text{ joined: } & A_0 = B_0 \cdot H \cdot B_1^{-1} \cdot H' \cdot B_2; \\ n \equiv 5, \text{ split: } & A_0 = B_0 \cdot H \cdot B_2, & \text{remnant circle } B_1^{-1} \cdot H'; \\ n \equiv 5, \text{ joined: } & A_0 = B_0 \cdot H \cdot B_1^{-1} \cdot H' \cdot B_2. \end{aligned}$$

For $n \equiv 1, 2 \pmod{6}$ the variety W is smooth, B meets no circle, and the arc is $A_0 = B$. In all cases let A be the arc component of L_1 , and let $\tilde{A}$ be the lift of A to $\mathbb{R}^2$ that starts at the lattice point $(0, 0)$ and enters $\{\gamma > 0\}$; its interior lies in $\mathbb{R}^2 \setminus (\pi\mathbb{Z})^2$. Let $\tilde{A}_0$ be the corresponding concatenation of lifts of the unperturbed pieces.

Lemma 6.1. *In each case of Table 1, $\tilde{A}$ joins $(0, 0)$ to $(\pi, 2\pi j)$ with the stated j , its interior lies in the open strip $0 < \gamma < \pi$, it is embedded, and φ along $\tilde{A}$ is piecewise monotone, taking the stated values at the ends of the pieces of $\tilde{A}_0$ up to $O(|\epsilon_A| + |\epsilon_B|)$. Near each junction at which the arrows in Table 1 change direction, φ has exactly one critical point on $\tilde{A}$; it has no other critical points. $\tilde{A}$ crosses exactly the lines Δ_k listed, each once and transversally, and meets no other line Δ_k except Δ_0 at its initial point.*

$n \bmod 6$	resol.	A_0	φ along $\tilde{A}_0$	j	lines crossed
1		B	$0 \nearrow \pi$	1	none
2		B	$0 \nearrow 3\pi$	2	Δ_1
4	split	$B_0 H B_2$	$0 \searrow -3\gamma_* \nearrow \pi + 3\gamma_* \searrow \pi$	1	Δ_0
4	joined	$B_0 H B_1^{-1} H' B_2$	$0 \searrow -3\gamma_* \nearrow \pi + 3\gamma_* \nearrow 4\pi - 3\gamma_* \nearrow 5\pi + 3\gamma_* \searrow 5\pi$	3	$\Delta_0, \Delta_1, \Delta_2$
5	split	$B_0 H B_2$	$0 \searrow -\gamma_* \nearrow 3\pi + \gamma_* \searrow 3\pi$	2	Δ_0, Δ_1
5	joined	$B_0 H B_1^{-1} H' B_2$	$0 \searrow -\gamma_* \nearrow 3\pi + \gamma_* \nearrow 4\pi - \gamma_* \nearrow 7\pi + \gamma_* \searrow 7\pi$	4	$\Delta_0, \dots, \Delta_3$

TABLE 1. The arc A in the limit $(\epsilon_A, \epsilon_B) \rightarrow 0$. The arrows describe φ along consecutive pieces; the values are those of φ at the ends of the pieces.

Proof. For $n \equiv 1, 2$, (F1) and (F4) give the first two rows: $\varphi = \gamma$ or 3γ along a segment from $(0, 0)$ to $(\pi, 2\pi)$ or $(\pi, 4\pi)$. Let $n \equiv 4$. By (F1) the lift of B_0 runs from $(0, 0)$ to $(\gamma_*, -2\gamma_*)$ with $\varphi = -3\gamma$. By (F2) the lift of H from there changes γ by $\pi - 2\gamma_*$ and φ by $\pi + 6\gamma_*$, hence θ by $2\pi + 4\gamma_*$, and ends at $(\pi - \gamma_*, 2\pi + 2\gamma_*)$. In the split case B_2 continues along a line of slope -2 to $(\pi, 2\pi)$, so φ decreases from $\pi + 3\gamma_*$ to π . In the joined case B_1^{-1} runs along the line $\theta = -2\gamma + 4\pi$ back to $(\gamma_*, 4\pi - 2\gamma_*)$, where $\varphi = 4\pi - 3\gamma_*$; then H' ends at $(\pi - \gamma_*, 6\pi + 2\gamma_*)$ and B_2 at $(\pi, 6\pi)$. The case $n \equiv 5$ is the same with $\varphi = -\gamma$ on B and the increment $3\pi + 2\gamma_*$ on each half; for example in the joined case the junctions lie at $(\gamma_*, 0)$, $(\pi - \gamma_*, 4\pi)$, $(\gamma_*, 4\pi)$, $(\pi - \gamma_*, 8\pi)$ and the end at $(\pi, 8\pi)$.

Since $0 < \gamma_* \leq \pi/6$, no value of φ at a junction lies in $2\pi\mathbb{Z}$. On each piece φ is strictly monotone (by (F1) on the pieces of B , which are straight with slope $\neq 1$, and by (F2)), so it crosses each line Δ_k with $2\pi k$ in the open range of the piece exactly once, transversally; reading off the ranges gives the last column. The ends: $\varphi(\pi, 2\pi j) = (2j - 1)\pi \notin 2\pi\mathbb{Z}$, and Δ_0 passes through $(0, 0)$.

Embeddedness of $\tilde{A}_0$ away from the junctions follows from the ranges of γ and φ on the pieces. On B_0 , $\gamma \leq \gamma_*$; on B_2 , $\gamma \geq \pi - \gamma_*$; on the halves, $\gamma_* < \gamma < \pi - \gamma_*$ in the interior by (F2); on B_1 , $\gamma_* \leq \gamma \leq \pi - \gamma_*$. Two pieces in the same γ -range have disjoint φ -ranges except at a common junction, as the values in Table 1 show; for instance in the joined case for $n \equiv 4$ the ranges on H , B_1^{-1} and H' are $[-3\gamma_*, \pi + 3\gamma_*]$, $[\pi + 3\gamma_*, 4\pi - 3\gamma_*]$ and $[4\pi - 3\gamma_*, 5\pi + 3\gamma_*]$, and φ is strictly monotone on each. Hence $\tilde{A}_0$ is injective. By (F4), $\tilde{A}$ is C^1 -close to $\tilde{A}_0$ away from the junctions and the ends, embedded near each junction, and embedded with $|d\varphi|$ bounded below near the ends. By compactness $\tilde{A}$ is embedded for small (ϵ_A, ϵ_B) , and the statements about crossings and monotonicity carry over. At a junction where the two pieces move φ in the same sense away from the junction point, which is where the arrows change direction, (F4) gives exactly one critical point; at the other junctions, and in the interiors of the pieces, there is none. $\square$

For the split resolution, the remnant circle has the same properties as a non-central circle.

Lemma 6.2. *The remnant circle misses the left and right edges, φ is strictly monotone along it with $d\varphi \neq 0$, and φ changes by $\mp 4\pi$ over one period ($-$ for $n \equiv 4$, $+$ for $n \equiv 5$). Its vertical degree is 2.*

Proof. For $n \equiv 4$ the circle is $B_1 \cdot H'^{-1}$. Along B_1 , $\varphi = -3\gamma + \text{const}$ decreases by $3(\pi - 2\gamma_*)$, and along H'^{-1} it decreases by $\pi + 6\gamma_*$, by (F2); the total is -4π . At both junctions the two pieces move φ in opposite senses away from the junction point, so (F4) gives no critical point there. The values of γ lie in $[\gamma_*, \pi - \gamma_*]$ up to $O(|\epsilon_A| + |\epsilon_B|)$, and γ returns to its initial value, so θ changes by -4π . For $n \equiv 5$ the circle is $B_1^{-1} \cdot H'$, and φ increases by $(\pi - 2\gamma_*) + (3\pi + 2\gamma_*) = 4\pi$. $\square$

6.2. Generators.

Proposition 6.3. *The curves L_0 and L_1 meet transversally. The arc carries $1 + 2c_A$ generators, where c_A is the number of lines Δ_k in the last column of Table 1: the generator r_+ and a pair $x_i^\pm$ over each crossing. Every non-central circle and the remnant circle carries four generators. In total there are $1 + |\sigma(K)|$ generators.*

Proof. The strands $S_\pm^k$ lie within $\sqrt{2}\epsilon$ of Δ_k and are C^1 -close to it (their slopes are $(1 \mp \epsilon \cos t)/(1 \pm \epsilon \cos t)$). By Lemma 6.1, outside small neighbourhoods of its crossings with the lines Δ_k and of $(0, 0)$, the curve $\tilde{A}$ stays a positive distance from $\bigcup_k \Delta_k$, so for small ϵ it meets no strand there. Near a transverse crossing with Δ_k , $\tilde{A}$ is a graph over φ with slope bounded away from 1, so it meets each of S_+^k and S_-^k exactly once and transversally; these are x^+ and x^- . Near $(0, 0)$, L_0 and the arc meet exactly once, at r_+ , by [HHK18, Lemma 5.1] and the corner chart lemma,

which bounds the limiting slope away from 1 and $|d\varphi|$ from below. The endpoint $(\pi, 2\pi j)$ lies on $\{\varphi = (2j - 1)\pi\}$, at distance about $\pi/\sqrt{2}$ from every Δ_k . The same argument applies to a circle: by (F3) and Lemma 6.2, φ advances monotonically by 4π over one period, so the circle crosses $\bigcup_k \Delta_k$ in exactly two points of P , transversally, giving four generators.

For the total, write $n = 6k + r$. There are $2\lfloor \nu/2 \rfloor = 2k$ non-central circles (Proposition 3.6). By Table 1 the arc carries 1, 3, 3 or 7, 5 or 9 generators for $r = 1, 2, 4, 5$ (split or joined), and the remnant circle adds 4 in the split cases. In every case the total is $8k + 1, 8k + 3, 8k + 7, 8k + 9$. On the other hand $|\sigma(T(3, n))| = 2(n - 1) - 4\lfloor n/6 \rfloor$ by Litherland's formula [Lit79] (in the form [Wue26c, Lemma 2.3]), which is $8k, 8k + 2, 8k + 6, 8k + 8$. $\square$

6.3. Bigons.

Lemma 6.4. *Let u be a bigon from p to q . Its boundary lifts to a closed curve in $\mathbb{R}^2 \setminus (\pi\mathbb{Z})^2$ consisting of a sub-path $\tilde{\beta}$ of one lift of a component of L_1 , from a lift $\tilde{q}$ of q to a lift $\tilde{p}$ of p , and a sub-arc of one strand S_σ^k from $\tilde{p}$ to $\tilde{q}$. If φ is strictly monotone along $\tilde{\beta}$, then no such bigon exists. In particular a component of L_1 along which φ is strictly monotone carries no bigon.*

Proof. The first statement holds because u is a map of a disc into P^* and $\mathbb{R}^2 \setminus (\pi\mathbb{Z})^2 \rightarrow P^*$ is a covering; the L_0 -side of the boundary is a path in the circle L_0 , whose lift lies in one component of the preimage of L_0 , a strand. Both $\tilde{p}$ and $\tilde{q}$ lie on S_σ^k , so $|\varphi(\tilde{p}) - 2\pi k|$ and $|\varphi(\tilde{q}) - 2\pi k|$ are at most 2ϵ . If φ is strictly monotone along $\tilde{\beta}$, then $\tilde{\beta}$ lies in the band $\{|\varphi - 2\pi k| \leq 2\epsilon\}$. For small ϵ the intersection of a lift of a component of L_1 with this band consists of short arcs, one around each transverse crossing with Δ_k (and, for the arc and $k = 0$, one at the corner $(0, 0)$), on each of which $|d\varphi|$ is bounded below in arclength: at the crossings because $d\varphi \neq 0$ there by Lemmas 6.1 and 6.2 and (F3), and at the corner by the corner chart lemma. Such a short arc meets S_σ^k only once, as in the proof of Proposition 6.3. Hence $\tilde{p} = \tilde{q}$, and the closed boundary curve would be the strand arc from $\tilde{p}$ to itself, which is constant: a contradiction. $\square$

For $n \equiv 4, 5 \pmod{6}$ let $x_1^\pm$ be the two generators over the crossing of $\tilde{A}$ with Δ_0 .

Proposition 6.5. *For $n \equiv 1, 2 \pmod{6}$ the complex has no bigons. For $n \equiv 4, 5 \pmod{6}$ it has exactly one bigon, from x_1^- to r_+ . In particular $\partial x_1^- = r_+$ and the differential vanishes on all other generators.*

Proof. There are no bigons between generators on different components [HHK18, p. 37], and none on a circle, by Lemma 6.4, (F3) and Lemma 6.2. For $n \equiv 1, 2$, φ is strictly monotone along $\tilde{A}$, so the arc carries none either.

Let $n \equiv 4, 5$. Since the arc is an interval, the L_1 -side of a bigon between two of its generators is the sub-path of $\tilde{A}$ between their lifts. By Lemma 6.4 it contains a critical point of φ , and by Lemma 6.1 there are exactly two: one near the end of B_0 , where φ attains the minimum value $\varphi_- = -3\gamma_*$ or $-\gamma_*$, and one near the start of B_2 , where it attains the maximum value $\varphi_+ \in \{\pi + 3\gamma_*, 5\pi + 3\gamma_*, 3\pi + \gamma_*, 7\pi + \gamma_*\}$. After the second critical point φ decreases from φ_+ to $\varphi_+ - 3\gamma_*$ or $\varphi_+ - \gamma_*$, an interval at distance at least $\pi/2$ from $2\pi\mathbb{Z}$, so no generator lies there; hence the sub-path contains only the first critical point, and one of its ends lies on the initial segment before it. On that segment $\varphi_- \leq \varphi \leq 0$ up to $O(|\epsilon_A| + |\epsilon_B|)$, so both ends lie on strands near Δ_0 , and the only generator on the initial segment is r_+ , which lies near the corner. The other end is a generator near Δ_0 after the minimum, that is x_1^+ or x_1^- . By the corner chart lemma the arc leaves $(0, 0)$ below Δ_0 , as B does; the strand S_-^0 passes $(0, 0)$ through $(\epsilon, -\epsilon)$ and S_+^0 through $(-\epsilon, \epsilon)$, so the first intersection r_+ of $\tilde{A}$ with a strand lies on S_-^0 . Since $x_1^- \in S_-^0$ and $x_1^+ \in S_+^0$, the only possible bigons join r_+ and x_1^- .

Existence. The strand S_-^0 is the graph of a function $\theta = g(\gamma)$, since $\gamma(t) = t + \frac{\pi}{2} - \epsilon \sin t$ is strictly increasing. Its intersections with $\tilde{A}$ are lifts of generators on L_0 near Δ_0 , so $\tilde{A} \cap S_-^0 = \{\tilde{r}_+, \tilde{x}_1^-\}$.

The lattice point $(0, 0)$ lies above S_-^0 , and $\tilde{A}$ crosses S_-^0 at $\tilde{r}_+$; hence the sub-path $\tilde{\beta}$ of $\tilde{A}$ from $\tilde{r}_+$ to $\tilde{x}_1^-$ lies below S_-^0 except at its ends. Let Λ be the simple closed curve formed by $\tilde{\beta}$ and the arc λ of S_-^0 from $\tilde{x}_1^-$ back to $\tilde{r}_+$. Both lie in the open strip $0 < \gamma < \pi$: $\tilde{\beta}$ by Lemma 6.1, and λ because γ is monotone along it between $\gamma(\tilde{r}_+) > 0$ and $\gamma(\tilde{x}_1^-) < \pi$. The two closed half-planes complementary to the strip are connected, unbounded and disjoint from Λ , so the bounded component D of $\mathbb{R}^2 \setminus \Lambda$ lies in the strip and contains no lattice point. Its corners are convex. At $\tilde{x}_1^-$ the continuation of $\tilde{A}$ beyond $\tilde{x}_1^-$ does not enter D , since it would have to leave D again across Λ , whereas $\tilde{A}$ is embedded, meets λ only at its ends, and ends at the lattice point $(\pi, 2\pi j) \notin \overline{D}$. At $\tilde{r}_+$ the initial piece of $\tilde{A}$ runs to the lattice point $(0, 0) \notin \overline{D}$, and the continuation of S_-^0 reaches $\{\gamma < 0\}$, both without meeting Λ again, so neither enters D . At each corner the two branches of Λ are adjacent among the four rays cut out by two transverse curves, and D does not contain the other two; so D occupies a single quarter, and the corner is convex. Near λ the domain D lies below S_-^0 , so its counterclockwise boundary traverses λ in the direction of decreasing γ , from $\tilde{x}_1^-$ to $\tilde{r}_+$, and then $\tilde{\beta}$ from $\tilde{r}_+$ to $\tilde{x}_1^-$. The projection of D to P^* is therefore a bigon from x_1^- to r_+ .

Uniqueness. The L_1 -side of a bigon between r_+ and x_1^- lifts to $\tilde{\beta}$, and its L_0 -side to an arc of S_-^0 that closes up with $\tilde{\beta}$; since S_-^0 is an embedded line, this arc is λ . A bigon from r_+ to x_1^- would have counterclockwise boundary running along λ from $\tilde{r}_+$ to $\tilde{x}_1^-$, so its boundary would wind -1 times around the points of D , which is impossible for an orientation-preserving immersion. By [HHK18, Cor. 3.13] there is at most one bigon from x_1^- to r_+ . $\square$

6.4. Restrictedness and admissibility.

Lemma 6.6. *L_1 is a restricted immersed 1-manifold, and (L_0, L_1) is an admissible pair.*

Proof. By Section 5 the arc is a proper immersion of an interval with limiting slopes $\neq 1$ at both ends. A fishtail on the arc would be a sub-path whose two ends map to the same point of P and which is nullhomotopic in P^* ; its lift would be a sub-path of $\tilde{A}$ with equal endpoints, contradicting Lemma 6.1. So the arc is a restricted immersed arc. For a circle component (non-central or remnant), a nullhomotopic loop lifts to a closed sub-path of a lift, along which φ would return to its initial value; by (F3) and Lemma 6.2 this is impossible, so there is no fishtail. The circle is essential, since its vertical degree is 2. The tangent line never equals ℓ_1 , because $d\varphi \neq 0$, so $\mu(L_1, \ell_1) = 0$; and a curve winding twice around the open strip has $z = \pm 4 \equiv 0$. Hence each circle is a restricted immersed circle (Definition 2.1).

For admissibility (Definition 2.2), L_0 is a circle and the intersections are transverse by Proposition 6.3. Loops in the domain of the arc are nullhomotopic. In $H_1(P^*; \mathbb{Z})$, let e_c be the class of a small counterclockwise loop around the corner c ; these classes generate, subject only to $\sum_c e_c = 0$. Since L_0 lies in a small neighbourhood of Δ , a disc containing only the corners $(0, 0)$ and (π, π) , every loop on L_0 has class in the span of $e_{(0,0)}$ and $e_{(\pi,\pi)}$. A circle of L_1 with vertical degree 2 missing the edges is homotopic in the open strip to twice the core of that annulus, whose class is $\pm(e_{(0,0)} + e_{(0,\pi)})$; a loop traversing it j times has class $\pm 2j(e_{(0,0)} + e_{(0,\pi)})$. If this lay in the span of $e_{(0,0)}$ and $e_{(\pi,\pi)}$ modulo $\sum_c e_c$, comparing the coefficients of $e_{(\pi,0)}$ and $e_{(0,\pi)}$ would give $j = 0$. Hence a loop on L_0 is freely homotopic to a loop on L_1 only if the latter is nullhomotopic, and then so is the former, since L_0 is essential and $\pi_1(P^*)$ is free. $\square$

6.5. Gradings. The grading is determined by two observations.

Lemma 6.7. *Let α be a path in L_1 along which φ is strictly monotone with $d\varphi \neq 0$. Then $\mu(\alpha, \ell_1) = 0$.*

Proof. A tangent vector $(\dot{\gamma}, \dot{\theta})$ with $\dot{\theta} - \dot{\gamma} = d\varphi(\dot{\gamma}, \dot{\theta}) \neq 0$ is not parallel to $(1, 1)$, so the tangent line never meets ℓ_1 , and μ counts no passages. $\square$

By Lemmas 6.1 and 6.2 and (F3), the hypothesis holds on every sub-path of L_1 that contains no critical point of φ . In particular it holds across the junctions of Table 1 at which φ is monotone, where by (F4) the velocity is a positive combination of two directions along which φ changes in the same sense.

Lemma 6.8. *Let p, q be generators lying on strands of the same sign $\sigma \in \{+, -\}$, with lifts $\tilde{p} \in S_\sigma^k$ and $\tilde{q} \in S_\sigma^{k \pm 1}$ joined by a sub-path of a lift of a component of L_1 that lies in the open strip $0 < \gamma < \pi$ and contains no critical point of φ . Then $\text{gr}(p, q) = 2$.*

Proof. We use (2.2), with α_1 the given sub-path from q to p and α_0 the branch of L_0 of slope < 1 (for $\sigma = +$) or > 1 (for $\sigma = -$) on the front face, from p to the point over q . Its lift from $\tilde{p}$ runs along S_σ^k to the point $\tilde{q} \mp (0, 2\pi)$, inside the open strip. By Lemma 6.1, $p = r_+$ satisfies the hypotheses only for $n \equiv 2 \pmod{6}$, the case used in Proposition 6.9; then $r_+ \in S_+^0$ lies at $\gamma \approx \frac{2}{3}\epsilon > 0$ on the side of $(0, 0)$ away from which S_+^0 leaves the strip, so this lift also stays in the open strip. Along α_0 the slope of the strand stays on one side of 1, so $\mu(\alpha_0, \ell_1) = 0$; $\mu(\alpha_1, \ell_1) = 0$ by Lemma 6.7; and the triple indices at p and q agree, since in both cases TL_0 is a line of slope just below (or just above) 1 and TL_1 a line of slope bounded away from 1. Finally, the lift of $\alpha_0 * \alpha_1$ runs in the open strip from $\tilde{q}$ to $\tilde{q} \mp (0, 2\pi)$. The open strip modulo the translation by $(0, 2\pi)$ is the annulus obtained from P by removing the two edges, so $\alpha_0 * \alpha_1$ is homotopic in P^* to $\pm$ the core of this annulus, which encircles two corners; thus $z(\alpha_0 * \alpha_1) = \pm 2 \equiv 2$. This is (2.3) with the right-hand side equal to 2. $\square$

Proposition 6.9. *Put $\text{deg} = \text{gr} - \sigma \pmod{4}$. The generators on the arc have the following degrees:*

case	degrees of r_+ ; (x_i^-, x_i^+)	HF of the arc, degrees $0, \dots, 3$
1	0	(1, 0, 0, 0)
2	0; (1, 2)	(1, 1, 1, 0)
4, split	0; (1, 2)	(0, 0, 1, 0)
4, joined	0; (1, 2), (3, 0), (1, 2)	(1, 1, 2, 1)
5, split	0; (1, 2), (3, 0)	(1, 0, 1, 1)
5, joined	0; (1, 2), (3, 0), (1, 2), (3, 0)	(2, 1, 2, 2)

Here $x_i^\pm$ are the generators over the i -th crossing along $\tilde{A}$. Each circle component contributes rank 1 in each degree, and the two generators x^- on a circle have degrees differing by 2.

Proof. $\text{deg } r_+ = 0$ by (2.4), and $\text{gr } x_i^+ = \text{gr } x_i^- + 1$ by [HHK18, §5.1]. For $n \equiv 2$, $\tilde{A}$ leaves $(0, 0)$ above Δ_0 , so $r_+ \in S_+^0$, and Lemma 6.8 applied to r_+ and $x_1^+ \in S_+^1$ gives $\text{deg } x_1^+ = -2 \equiv 2$. For $n \equiv 4, 5$ the bigon of Proposition 6.5 gives $\text{deg } x_1^- = 1$ by [HHK18, Prop. 3.8]. The later crossings follow from Lemma 6.8, since by Lemma 6.1 there is no critical point of φ between consecutive crossings: each step changes the degree of x^+ by 2. The homology of the arc is obtained from these degrees by removing x_1^- and r_+ in the cases $n \equiv 4, 5$. A circle carries four generators, $x^\pm$ over each of its two crossings with Δ (Proposition 6.3), and no bigon (Lemma 6.4, (F3) and Lemma 6.2), so its homology equals its complex. Lemma 6.8, applied to the two crossings of one period, shows that the two generators x^- differ by 2. Since $\text{gr } x^+ = \text{gr } x^- + 1$ at each crossing, the four generators lie in four different degrees, and the homology has rank 1 in each degree. $\square$

Theorem 6.10. *For $n \geq 4$ coprime to 3 and sufficiently small $(\epsilon_A, \epsilon_B) \in \mathcal{C}_a$, the arc A of L_1 is a restricted immersed arc from $(0, 0)$ to $(\pi, 0)$ whose lift $\tilde{A}$ ends at $(\pi, 2\pi j)$, with j , the generators and their degrees as in Table 1 and Proposition 6.9. The arc carries a bigon if and only if $n \equiv 4, 5$*

(mod 6), and then exactly one, from x_1^- to r_+ . The two resolutions of the double points c_+, c_- give the same generators, the same bigon and the same total graded homology: the joined arc for $n \equiv 4$ has homology $(1, 1, 2, 1) = (0, 0, 1, 0) + (1, 1, 1, 1)$, the split arc plus the remnant circle, and the joined arc for $n \equiv 5$ has homology $(2, 1, 2, 2) = (1, 0, 1, 1) + (1, 1, 1, 1)$.

Proof. This combines Lemmas 6.1 and 6.6 with Propositions 6.3, 6.5 and 6.9. $\square$

Remark 6.11. By [HHK18, Thm. 4.1] (Remark 2.4), the homology of the arc depends only on its homotopy class rel endpoints. Since $\bar{A}$ lies in the open strip and joins $(0, 0)$ to $(\pi, 2\pi j)$, it is homotopic rel endpoints in $\mathbb{R}^2 \setminus (\pi\mathbb{Z})^2$ to the straight segment between these points, whose interior misses the lattice. Along that segment $\varphi = (2j - 1)\gamma$ is monotone and crosses $\Delta_1, \dots, \Delta_{j-1}$, so by Proposition 6.3 and Lemma 6.4 its complex has $2j - 1$ generators and no differential, in agreement with [HHK18, Thm. 7.1]. The values $2j - 1 = 1, 3, 1, 5, 3, 7$ for the six rows of Table 1 agree with Proposition 6.9. We do not use this in the proofs.

Remark 6.12. We have computed the perturbed curves numerically for the twelve values $n = 4, 5, 7, 8, 10, 11, 13, 14, 16, 17, 22, 23$, with $(n\epsilon_A, \epsilon_B, \epsilon) = (0.3, 0.05, 0.01)$ and $(-0.3, 0.03, 0.02)$, and evaluated (2.2) directly for every pair of generators on a component. In all 24 cases the generators, the bigon and the degrees agree with Theorem 6.10, and for $n = 4, 5, 7$ the complexes agree with those of [HHK18, §§11.2, 11.6, 11.7]. For the first perturbation we also checked that the tangent of L_1 is parallel to ℓ_1 only at the critical points of φ . A computation of the generator counts, bigons and homology ranks for all $4 \leq n \leq 50$ coprime to 3, at four perturbations each (128 cases), also agrees. In all these computations the resolution was split for $n \equiv 4$ and joined for $n \equiv 5$. For $n \equiv 5$ these perturbations lie in the sector containing the ϵ_A -axis. Near the ϵ_B -axis the split resolution occurs (Remark 5.8), and at the perturbations listed there the generators, the bigon and the degrees agree with the split rows of Table 1 and Proposition 6.9. The proofs do not need to know which resolution occurs.

6.6. Proofs of the main results. Write $n = 6k + r$ with $r \in \{1, 2, 4, 5\}$. By Proposition 6.9, the homology in degrees 0, 1, 2, 3 is the homology of the arc plus $(1, 1, 1, 1)$ for each circle; the number of circles is $2k$, plus one remnant circle in the split cases. Table 2 compares the result with (2.5), using $|\sigma| = 8k, 8k + 2, 8k + 6, 8k + 8$ (Proposition 6.3), $N_1 = \lceil |\sigma|/4 \rceil$, $N_3 = \lfloor |\sigma|/4 \rfloor$ and $R = \frac{1}{2}(1 + |\sigma| - \|\Delta_K\|_1)$. Here $\|\Delta_{T(3,n)}\|_1 = (4n - \eta)/3$, which equals $8k + 1, 8k + 3, 8k + 5, 8k + 7$, by Moree's formula $\|\Delta_{T(p,q)}\|_1 = 2ab - 1$ for $ap + bq = pq + 1$ [Mor14, Lemma 1, Cor. 1]; thus $R = 0, 0, 1, 1$.

r	$H^\natural$ in degrees 0, 1, 2, 3	(N_1, N_3)	R	(2.5)
1	$(2k + 1, 2k, 2k, 2k)$	$(2k, 2k)$	0	$(2k + 1, 2k, 2k, 2k)$
2	$(2k + 1, 2k + 1, 2k + 1, 2k)$	$(2k + 1, 2k)$	0	$(2k + 1, 2k + 1, 2k + 1, 2k)$
4	$(2k + 1, 2k + 1, 2k + 2, 2k + 1)$	$(2k + 2, 2k + 1)$	1	$(2k + 1, 2k + 1, 2k + 2, 2k + 1)$
5	$(2k + 2, 2k + 1, 2k + 2, 2k + 2)$	$(2k + 2, 2k + 2)$	1	$(2k + 2, 2k + 1, 2k + 2, 2k + 2)$

TABLE 2. The graded pillowcase homology of $T(3, 6k + r)$, in degrees $\text{gr} - \sigma$, and the graded rank of $I^\natural(T(3, 6k + r); \mathbb{Q})$ from (2.5). For $r = 4, 5$ both resolutions give the same vector.

Proof of Theorem 1.1. Part (1) is Lemma 6.6 and Proposition 6.3, and part (2) is Proposition 6.5. For part (3), Table 2 shows that $H^\natural(Y, T, \pi)$ has rank $\|\Delta_K\|_1$ and that, with $\text{gr } r_+ = \sigma$, its rank in grading g equals the rank of $I^\natural(K; \mathbb{Q})$ in degree $g - \sigma$ of (2.5), that is, in Kronheimer and Mrowka's grading g . Each circle contributes rank 1 in each grading, so this does not depend on how the absolute grading of the circle components is fixed. $\square$

Proof of Corollary 1.2. By Lemma 6.6, L_1 is a restricted immersed 1-manifold for every sufficiently small perturbation in the closed cone $\mathcal{C}_a$ (Definition 5.5), and by Theorem 1.1(3) the $\mathbb{F}_2$ -dimension of $H^\natural(Y, T, \pi)$ in each grading equals the $\mathbb{Q}$ -dimension of $I^\natural(K; \mathbb{Q})$ in the same grading; this is the form in which [HHK18, §11] test Conjecture 2.5. For $n \equiv 1, 2 \pmod{6}$ the S-complex $\tilde{C}(K; \mathbb{Z})$ has zero differential (see the proof of Theorem 1.3), so its homology, which is $I^\natural(K; \mathbb{Z})$ up to a shift of grading [DS24b, Thm. 8.9], is free; by the universal coefficient theorem $I^\natural(K; \mathbb{F}_2)$ then has the same graded ranks, and $H^\natural(Y, T, \pi) \cong I^\natural(K; \mathbb{F}_2)$ as graded vector spaces. $\square$

For the S-structure we fix the absolute grading of the circle components.

Definition 6.13. We grade each circle component of L_1 so that its two generators x^- have odd degree.

By Proposition 6.9 this is possible, and it determines the grading of the component up to a shift by 2, which exchanges the degrees of the two generators x^- (and of the two x^+); since each circle has one generator x^- in each odd degree either way, this ambiguity does not affect the statements below. The normalization agrees with the convention of [HHK18, §6.1], which gives one generator on each circle the grading of the corresponding generator of Kronheimer and Mrowka's complex, provided: (i) the identification of the relative grading of (2.2) with Kronheimer and Mrowka's grading, outlined in [HHK18, §6.1], under which x^- and x^+ correspond to the lower and upper of the two generators over one irreducible traceless character; (ii) the irreducible generators of $\tilde{C}(K)$ lie in odd degrees, $C_*(K) = C_1 \oplus C_3$ [DS24a, §1, Cor. 3]; (iii) Kronheimer and Mrowka's grading is the degree of $\tilde{C}(K)$ plus σ [Wue26c], [DS24b, Thm. 8.9]. Input (i) is only outlined in [HHK18].

Proof of Theorem 1.3. By Proposition 6.5 every bigon runs from x_1^- to r_+ , so Conjecture 2.6(a) holds and $\mathcal{P}(K)$ is an S-complex with $\chi(x_i^-) = x_i^+$. With the grading of Definition 6.13, the generators x^- have odd degree: on the arc by Proposition 6.9, on circles by definition. Counting them by degree, using Proposition 6.9 and $2k$ (plus one in the split case) circles, gives $2k$ and $2k$ generators x^- in degrees 1 and 3 for $r = 1$; $2k + 1$ and $2k$ for $r = 2$; $2k + 2$ and $2k + 1$ for $r = 4$; and $2k + 2$ and $2k + 2$ for $r = 5$, in both resolutions. These are (N_1, N_3) . Hence the map sending the generators x^- of degree 1 and 3 to bases of C_1 and C_3 , each $x^+ = \chi(x^-)$ to the corresponding copy, and r_+ to the reducible generator, is a degree-preserving isomorphism of graded vector spaces that commutes with χ . Under it the differential of $\mathcal{P}(K)$ is zero for $n \equiv 1, 2$, and for $n \equiv 4, 5$ it is the map $x_1^- \mapsto r_+$, a component from C_1 to the reducible summand of rank one: a map of the type of δ_1 .

Let $n \equiv 1, 2 \pmod{6}$. Then $R = 0$, so the differential of $\tilde{C}(K; \mathbb{Q})$ vanishes. The S-complex $\tilde{C}(K; \mathbb{Z})$ built on the character variety is a complex of free abelian groups whose differential is an integer matrix; it vanishes over $\mathbb{Q}$, hence is zero, and so is the differential of $\tilde{C}(K; \mathbb{F}_2) = \tilde{C}(K; \mathbb{Z}) \otimes \mathbb{F}_2$. Both $\mathcal{P}(K)$ (shifted by σ) and $\tilde{C}(K; \mathbb{F}_2)$ are then S-complexes with zero differential on the same graded generators, and the isomorphism above is an S-isomorphism. This proves Conjecture 2.6(b) for these n .

Let $n \equiv 4, 5 \pmod{6}$. By the above, $\mathcal{P}(K)$ shifted by σ is S-isomorphic to the S-complex on the graded generators of $\tilde{C}(K)$ whose only nonzero component is a rank-one map δ_1 from C_1 to the reducible summand. Hence Conjecture 2.6(b) for K is equivalent to the statement that $\tilde{C}(K; \mathbb{F}_2)$ is S-chain homotopy equivalent to this complex. We prove it. Recall that $d = 0$ and $\delta_2 = 0$, and write $v_{31}: C_3 \rightarrow C_1$ and $v_{13}: C_1 \rightarrow C_3$ for the components of v . Over $\mathbb{Q}$ the differential of $\tilde{C}(K)$ has rank $R = 1$ and $\delta_1 \neq 0$ [Wue26c], [DS24b, §9.5]. The map v_{31} and the

pair (v_{13}, δ_1) have sources and targets in different degrees, so $v_{31} = 0$ and $v_{13} = \delta_1(\cdot)u$ for some $u \in C_3 \otimes \mathbb{Q}$; since v_{31} is an integer matrix, it vanishes over $\mathbb{Z}$.

Next, δ_1 takes an odd value on $C_1(K; \mathbb{Z})$. Daemi and Scaduto define their invariant h for S-complexes over any integral domain that is an algebra over $\mathbb{Z}[U^{\pm 1}, T^{\pm 1}]$, in particular over $\mathbb{F}_2$ with $T = U = 1$ [DS24a, §2.2], and $\tilde{C}(K; \mathbb{F}_2) = \tilde{C}(K; \mathbb{Z}) \otimes \mathbb{F}_2$. Their cobordism inequality [DS24a, Thm. 4.22, Prop. 4.30] applies to the full twist on three strands, which carries $T(3, q)$ to $T(3, q+3)$; there the minimizing reducible is unique, so its count is a unit in $\mathbb{F}_2$ [DS24a, Cor. 4.8]. Starting from the unknot and $T(2, 3) = T(3, 2)$, for which $h \geq 0$ over every such ring [DS24a, §4], induction on q gives $h_{\mathbb{F}_2}(T(3, n)) \geq 1$ for $n \equiv 4, 5 \pmod{6}$. Since $d = 0$, this means that $\delta_1 \otimes \mathbb{F}_2 \neq 0$ [DS24b, Prop. 4.15, Rem. 4.2].

Write $\delta_1 = e\delta'_1$ with $\delta'_1: C_1(K; \mathbb{Z}) \rightarrow \mathbb{Z}$ surjective; then e is odd. For x_0 with $\delta'_1(x_0) = 1$ we have $eu = v_{13}(x_0) \in C_3(K; \mathbb{Z})$, so over $\mathbb{F}_2$ one has $v_{13} = \delta_1(\cdot)w$ with w the reduction of eu . The map $(\alpha, \beta, m) \mapsto (\alpha, \beta + mw, m)$ on $\tilde{C}(K; \mathbb{F}_2) = C_* \oplus C_{*-1} \oplus \mathbb{F}_2$ commutes with χ and is the identity on the reducible summand, so it is an S-isomorphism, and conjugating the differential by it replaces v_{13} by $v_{13} + \delta_1(\cdot)w = 0$. Hence $\tilde{C}(K; \mathbb{F}_2)$ is S-isomorphic to the complex whose only nonzero component is $\delta_1 \otimes \mathbb{F}_2$, of rank one. Two such complexes on the same graded generators are S-isomorphic, by a change of basis of C_1 applied to both copies. This proves Conjecture 2.6(b) for these n . $\square$

7. A SECOND DECOMPOSITION OF $T(3, 4)$

For a torus knot $T(p, q)$, every pair (r, s) with $pr + qs = 1$ defines a decomposition of the form used in Section 3, and different choices give different character varieties $R(Y, T)$ and different curves L_1 [HHK18, §11]. For $T(3, 4)$ the choices are $(r, s) = (3 + 4j, -2 - 3j)$, $j \in \mathbb{Z}$. Theorem 1.1 concerns $j = 0$, the choice of [HHK18, §11.6]. In this section we describe the complex for $j = 1$, that is $(r, s) = (7, -5)$. With zero bounding cochain the conclusion of Theorem 1.1(3) fails there. A nonzero bounding cochain on the arc restores it.

The results of this section were obtained by numerical computation in floating-point arithmetic, and they are not proofs. We traced the perturbed variety from HHK's equations (3.1) by numerical continuation, for 36 small perturbations (ϵ_A, ϵ_B) in twelve directions with $\epsilon_B \neq 0$, of sizes down to 0.003, and for earring parameters ϵ between 10^{-4} and 0.05. All statements below held in every case. As a control, the same computations for $j = 0$ reproduce the complex of Theorem 1.1. For $j = -1$, that is $(r, s) = (-1, 1)$, the curve is different, but the homology again has the graded dimensions of $I^{\natural}(T(3, 4))$.

Proposition 7.1 (numerical). *For $T(3, 4)$ with $(r, s) = (7, -5)$ and the perturbations above:*

- (1) $R_{\pi}(Y, T)$ consists of an arc and five circles. The lift of the arc starting at $(0, 0)$ ends at $(7\pi, 8\pi)$. In particular the arc is not homotopic relative to its ends to the arc of Theorem 1.1, whose lift ends at $(\pi, 2\pi)$.
- (2) L_1 is a restricted immersed 1-manifold and (L_0, L_1) is an admissible pair.
- (3) The complex has $7 = 1 + |\sigma|$ generators. The arc carries r_+ , x^- and x^+ , in gradings 2, 3 and 0. One circle carries four generators, one in each grading, and the other four circles are disjoint from L_0 .
- (4) With zero bounding cochain there is no bigon. Hence $H^{\natural}(Y, T, \pi)$ has dimensions $(2, 1, 2, 2)$ in gradings 0, 1, 2, 3, and rank 7, whereas $I^{\natural}(T(3, 4))$ has dimensions $(2, 1, 1, 1)$ and rank 5.
- (5) The arc has a self-intersection s near the corner $(0, 0)$ such that x^- , r_+ and s are the corners of an embedded triangle with convex corners. The point s is a bounding cochain of degree 1 for the arc, and in the complex deformed by it the differential is $x^- \mapsto r_+$. The deformed homology has dimensions $(2, 1, 1, 1)$.

We describe how each part was established.

The arc and its homotopy class. The lines $\varphi \in \pi\mathbb{Z}$ cut the plane into strips containing no lattice point. The lifted arc starts at $(0,0)$ in the strip $-\pi < \varphi < 0$. It crosses Δ_0 once, at $\gamma \approx 4.33\pi$, and then the line $\varphi = \pi$ once, at $\gamma \approx 2.67\pi$, before ending at $(7\pi, 8\pi)$. Since each strip is convex, the arc is homotopic relative to its ends in $\mathbb{R}^2 \setminus (\pi\mathbb{Z})^2$ to the polygon with vertices

$$(0,0), (0.5,0), (4.8,4.3), (4.8,5.3), (2.5,3), (2.5,4), (6.5,8), (7,8),$$

in units of π . This polygon meets L_0 in exactly three points for every $\epsilon \in [5 \cdot 10^{-4}, 0.2]$, and it bounds no bigon with L_0 . By [HHK18, Thm. 4.1] (Remark 2.4), the arc therefore contributes rank 3 independently of the traced geometry.

Gradings. The gradings in (3) were computed from formula (2.2). The Maslov index was evaluated by the floor formula, the triple indices from the tangent lines, and the class z from the homotopy class of the loop. Each value was computed for both directions of the path in L_0 , with the same result. With the same code, for $j = 0$ the arc generators have gradings 2, 3, 0 and the bigon of Theorem 1.1 is present.

No bigon with zero bounding cochain. As in Lemma 6.4, a bigon lifts to an immersed disk in $\mathbb{R}^2 \setminus (\pi\mathbb{Z})^2$, whose boundary on L_0 runs along a single lift of L_0 . Lifts of the two families S_+^k and S_-^k meet only over the double point of L_0 .

- On the arc, r_+ and x^- lie on the same lift S_-^0 , and x^+ lies on S_+^0 . Gradings alone allow a bigon from x^- to r_+ . Its boundary would be the sub-arc of the lifted arc from $\tilde{r}_+$ to $\tilde{x}^-$ together with the segment of S_-^0 between them. This closed curve has winding number 1 about the lattice points (π, π) and $(3\pi, 3\pi)$. The boundary of an immersed disk has winding number zero about every point outside its image, so there is no such bigon. For $j = 0$ the corresponding curve has winding number zero about every lattice point, and the bigon exists.
- On the circle with four generators, they lie on four distinct lifts of L_0 , also after translation by the period of the circle. So no bigon has both corners on that circle.

Restrictedness and admissibility. Every lift of every component of L_1 to the plane is embedded, so no component has an immersed monogon, and all components are unobstructed. The circles satisfy $\mu(\cdot, \ell_1) + z \equiv 0 \pmod{4}$. Four of the circles lift to curves with displacement $(4\pi, 4\pi)$ per period, contained in the band $-2.4 < \varphi < -0.75$. Such a curve passes every lattice point of the line $\varphi = 0$ on the same side, and every lattice point of the line $\varphi = -\pi$ on the other. It is therefore freely homotopic in P^* to the square of the simple closed curve of slope one that separates the corners $(0,0)$, (π, π) from the other two. The earring L_0 passes the lattice points of the line $\varphi = 0$ alternately on the two sides, so it represents the figure-eight class. No nonzero power of this class is conjugate to that square in the free group $\pi_1(P^*)$; the two are already different in $H_1(P^*)$. Hence condition (ii) of Definition 2.2 holds. These four circles are disjoint from L_0 and contribute nothing. The fifth circle, which carries the four generators, lifts to a curve with displacement $(0, \pm 4\pi)$ per period. For it, as for the other four circles, we checked that its class in $H_1(P^*)$ and that of L_0 are linearly independent, which gives condition (ii) as in the proof of Lemma 6.6.

The bounding cochain. The arc crosses itself in P twelve times, while the arc of Theorem 1.1 for $T(3,4)$ is embedded.

- One self-intersection s lies at $(\gamma, \theta) \approx (0.18\pi, 0.10\pi)$, inside the region near the corner $(0,0)$ that carries the bigon of Theorem 1.1 for $j = 0$.
- Through s , a translate of the branch of the arc that crosses Δ meets the initial branch of the arc. Together with the lift of L_0 through $\tilde{r}_+$ and $\tilde{x}^-$, these two branches bound an

embedded triangle with corners x^- , r_+ and s . Its corners are convex and it contains no lattice point.

- In the framework of bounding cochains for immersed curves used in [CHKK22], the element $b = s$ satisfies the Maurer–Cartan equation, and its degree is forced to be 1. The relevant polygons are those with all their corners on L_1 at s , and every candidate encloses a lattice point.
- The triangle then contributes $x^- \mapsto r_+$ to the deformed differential, which gives the dimensions $(2, 1, 1, 1)$.

Remark 7.2. The two tangles are different. The quotients of $\pi_1(Y \setminus T)$ by the meridians of the two strands identify them with the groups of the trefoil and of $T(5, 7)$ for $(r, s) = (7, -5)$, and with those of the unknot and of the trefoil for $(r, s) = (3, -2)$. So Proposition 7.1 contradicts no invariance property. Since Conjecture 6.5 of [HHK18] asserts only the existence of a suitable decomposition, it does not contradict that conjecture either. The mechanism in part (5) matches the one found in [CHKK22, §10.3] for $T(4, 5)$ with the earring on the other tangle, where a bounding cochain restores the rank of $I^\natural$. Here it occurs with the earring on the trivial tangle, within the family of [HHK18, §10]. Conjecture 2.6 concerns zero bounding cochains and the normalization $0 < r < q$, which $(r, s) = (7, -5)$ violates. This example shows, numerically, that the normalization $0 < r < q$ cannot be dropped while the bounding cochains are kept zero.

APPENDIX A. INTERVAL-ARITHMETIC VERIFICATIONS

This appendix proves the three inequalities used in Section 4. Each is an inequality between explicit real-analytic functions of two variables on a compact domain, and each is reduced to finitely many evaluations in interval arithmetic [MKC09]: every elementary function is evaluated on a box by a routine that returns an interval containing all its values, with outward rounding, so that a strictly positive lower endpoint proves positivity on the box. We use the interval context of `mpmath` [mpm23] at a working precision of 30 decimal digits, with π enclosed in an interval.¹ Where a function has a removable singularity we use the following enclosure of $\text{sinc } y = \sin y / y$ on an interval $[y_0, y_1] \subset [0, \pi]$: sinc is decreasing there, so its range is $[\text{sinc } y_1, \text{sinc } y_0]$, and each endpoint is evaluated either directly or, for $y < 10^{-6}$, by $1 - y^2/6 \leq \text{sinc } y \leq 1 - y^2/6 + y^4/120$.

A.1. The inequality for the non-central circles.

Lemma A.1. *On the closed region $\Omega_+ = \{0 \leq \delta \leq \frac{\pi}{2}, \frac{\pi}{4} - \frac{\delta}{2} \leq \psi \leq \frac{3\pi}{4} - \frac{\delta}{2}\}$,*

$$F_0 - 0.851 |F_1| \geq 1.648.$$

The same bound holds on $\Omega_- = \{-\frac{\pi}{2} \leq \delta \leq 0, \frac{\pi}{4} - \frac{\delta}{2} \leq \psi \leq \frac{3\pi}{4} - \frac{\delta}{2}\}$.

Proof. Parametrize Ω_+ by $(t, \delta) \in [0, 1] \times [0, \frac{\pi}{2}]$, $\psi = \frac{\pi}{4} - \frac{\delta}{2} + \frac{\pi}{2}t$; this map is a bijection onto Ω_+ . Subdivide $[0, 1] \times [0, \frac{\pi}{2}]$ into $96 \times 96 = 9216$ boxes. On each box evaluate F_0 and F_1 in interval arithmetic, with ψ and δ as interval inputs, and take the lower endpoint of F_0 minus 0.851 times the larger endpoint of $|F_1|$. The smallest of these 9216 numbers is 1.6482..., and none is negative. The statement for Ω_- follows from the invariance of F_0 and F_1 under $(\psi, \delta) \mapsto (\pi - \psi, -\delta)$, which maps Ω_- onto Ω_+ .

As an independent check we also used Lipschitz bounds. With the same (t, δ) coordinates on either region one has $|\partial_t F_0| \leq 12\pi$, $|\partial_\delta F_0| \leq 8$, $|\partial_t F_1| \leq 6\pi$ and $|\partial_\delta F_1| \leq 10$, obtained by differentiating term by term. So $f = F_0 - 0.851 |F_1|$ satisfies $|f(p) - f(q)| \leq (12\pi + 6\pi \cdot 0.851) |\Delta t| +$

¹The three computations are carried out by short Python scripts built on `mpmath`, which are available, with the computations of Sections 6 and 7, at <https://github.com/bwuebben/pillowcase-bounding-cochain>. Each runs in under a minute on a laptop.

$(8 + 10 \cdot 0.851)|\Delta\delta|$. Evaluating f in interval arithmetic at the centres of a 240×240 grid on each of Ω_+ and Ω_- and subtracting the Lipschitz slack gives $f \geq 1.637$ on both regions. $\square$

The constant 0.851 is chosen just above $1/(2 \sin(\pi/5)) = 0.850651\dots$, the bound for $|\lambda|$ in the proof of Theorem 4.4; a float evaluation on a fine grid shows that $F_0 - c|F_1| > 0$ on Ω_+ for every $c < 2$, so the margin is large.

A.2. The inequality for the central circle. For $n \equiv 4$ let $D_4 = \{(\xi, h) : 0 \leq h \leq 1, 0 < \xi \leq \pi/(2+h)\}$, and for $n \equiv 5$ let $D_5 = \{(\xi, h) : 0 \leq h \leq \frac{1}{2}, 0 < \xi \leq \pi/(2-h)\}$; the parameters of Section 4.3 lie in these domains. The functions $\mathcal{E}_4, \mathcal{E}_5$ are those of Lemma 4.9.

Lemma A.2. *For $j = 4, 5$, the function $\mathcal{E}_j(\xi, h)/\xi^3$ is positive on D_j . More precisely, $\mathcal{E}_j$ is odd in ξ ,*

$$\mathcal{E}_j = c_3\xi^3 + c_5\xi^5 + c_7\xi^7 + O(\xi^9),$$

with

$$\begin{aligned} n \equiv 4 : \quad c_3 &= 4(h+1)(h+2)(h+3), \quad c_5 = -\frac{2}{5}(h+1)(h+2)(h+3)(5h^2 + 8h + 12), \\ c_7 &= \frac{1}{630}(h+1)(h+2)(h+3)(273h^4 + 960h^3 + 2080h^2 + 2160h + 1520); \\ n \equiv 5 : \quad c_3 &= \frac{4}{3}(2-h)(11-2h), \quad c_5 = \frac{2}{15}(2-h)(18h^3 - 99h^2 + 152h - 116), \\ c_7 &= \frac{1}{630}(h-2)(502h^5 - 3497h^4 + 9456h^3 - 14128h^2 + 11504h - 4656), \end{aligned}$$

and one has $\mathcal{E}_4/\xi^3 \geq 22.06$, $\mathcal{E}_5/\xi^3 \geq 19.18$ for $\xi \leq \frac{1}{4}$, and $\mathcal{E}_4/\xi^3 \geq 0.798$, $\mathcal{E}_5/\xi^3 \geq 0.650$ for $\xi \geq \frac{1}{4}$.

Proof. The region $\xi \leq \frac{1}{4}$. The Taylor coefficients are computed symbolically; the even ones vanish because $\mathcal{E}_j$ is odd. By Taylor's theorem $\mathcal{E}_j = c_3\xi^3 + c_5\xi^5 + c_7\xi^7 + R$ with $|R| \leq M_9\xi^9/9!$, where M_9 bounds $|\partial_\xi^9 \mathcal{E}_j|$ on $[0, \frac{1}{4}] \times [0, h_{\max}]$. To bound M_9 , write $\mathcal{E}_j = Sg + q$ with $S = \sin(\xi h)/h$ and g, q trigonometric polynomials in ξ whose frequencies are linear in h . By the product-to-sum formulas,

$$\begin{aligned} n \equiv 4 : \quad g &= 6 + 5 \cos 2\xi - 3 \cos(2+2h)\xi - 2 \cos(4+2h)\xi, \\ q &= 2 \sin h\xi + \sin(2+h)\xi - \sin(2-h)\xi \\ &\quad - \frac{1}{2} [\sin(4+3h)\xi + \sin(2+h)\xi + \sin(4+h)\xi + \sin(2+3h)\xi], \\ n \equiv 5 : \quad g &= 6 - 5 \cos 2\xi + 3 \cos(2-2h)\xi - 2 \cos(4-2h)\xi, \\ q &= -2 \sin h\xi + \sin(2+h)\xi - \sin(2-h)\xi \\ &\quad - \frac{1}{2} \sin(4-h)\xi + \frac{1}{2} \sin(2-3h)\xi - \frac{1}{2} \sin(4-3h)\xi + \frac{1}{2} \sin(2-h)\xi. \end{aligned}$$

Since $|S| \leq \xi$ and $|\partial_\xi^j S| \leq h^{j-1}$ for $j \geq 1$, the Leibniz rule gives

$$M_9 \leq \xi|g^{(9)}| + \sum_{j \geq 1} \binom{9}{j} h^{j-1} |g^{(9-j)}| + |q^{(9)}|,$$

and each derivative of a trigonometric polynomial $\sum a_i \cos \omega_i \xi$ is bounded by $\sum |a_i| |\omega_i|^k$. Bounding h^{j-1} by $h_{\max}^{j-1}$ and each $|\omega_i|$ by its larger value at $h = 0$ or $h = h_{\max}$ gives $M_9 \leq 9.32 \cdot 10^7$ for $n \equiv 4$ and $M_9 \leq 2.44 \cdot 10^6$ for $n \equiv 5$, hence $|R|/\xi^3 \leq 0.063$, resp. 0.0017, on $\xi \leq \frac{1}{4}$. Evaluating $c_3 + c_5\xi^2 + c_7\xi^4$ in interval arithmetic on $\xi \in [0, \frac{1}{4}]$ and 64 subintervals of h gives the stated lower bounds.

The region $\xi \geq \frac{1}{4}$. Cover the part of D_j with $\xi \geq \frac{1}{4}$ by $64 \times 64 = 4096$ boxes (for each of 64 subintervals of h , the interval of ξ from $\frac{1}{4}$ to the larger of the two endpoint values of $\pi/(2 \pm h)$ is divided into 64 parts). On each box evaluate $\mathcal{E}_j/\xi^3$ in interval arithmetic, writing

$\sin(\xi h)/h = \xi \operatorname{sinc}(\xi h)$ with the enclosure above; here $\xi h \leq \pi/2$. All 4096 lower endpoints are positive, with minimum 0.798, resp. 0.650. $\square$

As an independent check, the same inequality was verified with a different splitting point $\xi = 0.2$ and adaptive subdivision of the boxes; this gives $\mathcal{E}_4/\xi^3 \geq 22.80$ and $\mathcal{E}_5/\xi^3 \geq 19.48$ near $\xi = 0$, and positive lower bounds on all boxes.

A.3. The inequality for γ on the central circle. Let $\mathcal{N}_4$ and $\mathcal{N}_5$ be the functions in the proof of Proposition 4.11, and put $p_4 = 3$, $p_5 = 4$.

Lemma A.3. *For $j = 4, 5$, $\mathcal{N}_j(\xi, 0) = 0$ identically, and $\mathcal{N}_j/(h\xi^{p_j})$ extends to a real-analytic function on D_j , including $h = 0$, which is positive at every point of D_j except the point $(\xi, h) = (\pi/2, 0)$ of D_4 , where it vanishes; in particular it is positive wherever $h > 0$. Its value at $\xi = 0$ is*

$$\frac{4}{3}(h+1)(h+2)(3h+1) \quad (j=4), \quad \frac{2}{3}(2-h)(2h+1) \quad (j=5).$$

Proof. The factor h . With $\alpha = \xi h$, the sum-to-product formulas give the identities

$$\begin{aligned} \mathcal{N}_4 &= 2h \cos \xi \cdot \operatorname{Br}_4, & \operatorname{Br}_4 &= (2+h) \xi \operatorname{sinc} \alpha - \sin(2\xi + \alpha) + 2(1+h) h \xi^2 \operatorname{sinc}^2 \alpha \sin(2\xi + \alpha), \\ \mathcal{N}_5 &= h \sin \xi \cdot \operatorname{Br}_5, & \operatorname{Br}_5 &= (2-h) \xi \operatorname{sinc} \alpha - \sin(2\xi - \alpha) + 2h \xi^2 \operatorname{sinc}^2 \alpha \sin(2\xi - \alpha). \end{aligned}$$

For instance, for $j = 4$: $\sin(\xi + \alpha) + \sin(3\xi + \alpha) = 2 \sin(2\xi + \alpha) \cos \xi$ and $2 \sin(\xi + \alpha) \cos \xi = \sin(2\xi + \alpha) + \sin \alpha$, so $\mathcal{N}_4/(4 \cos \xi) = \sin(2\xi + \alpha) [-\frac{h}{2} + (1+h) \sin^2 \alpha] + \frac{2+h}{2} \sin \alpha$, and $\sin \alpha = h \xi \operatorname{sinc} \alpha$. On D_4 we have $\xi < \pi/2$, so $\cos \xi > 0$, except at the point $(\pi/2, 0)$, where $\cos \xi = 0$; on D_5 we have $\xi < \pi$, so $\sin \xi > 0$. Hence, away from that point, $\mathcal{N}_j/(h\xi^{p_j}) > 0$ if and only if $\operatorname{Br}_j/\xi^{p_j} > 0$.

The region $\xi \leq 0.3$. The Taylor coefficients of $\mathcal{N}_j/h$ in ξ are polynomials in h with rational coefficients, computed symbolically; those below ξ^{p_j} vanish, and the leading one is the stated value. Keep the terms of degree $p_j, \dots, p_j + 5$. For the remainder, write $\mathcal{N}_j/h = \int_0^1 \partial_h \mathcal{N}_j(\xi, th) dt$, so that $\partial_\xi^K (\mathcal{N}_j/h)$ lies in the convex hull of the values of $\partial_\xi^K \partial_h \mathcal{N}_j$ on $[0, 0.3] \times [0, h]$; this hull is enclosed in interval arithmetic for $K = p_j + 6$. On 64 subintervals of h this gives $\mathcal{N}_4/(h\xi^3) \geq 2.48$ and $\mathcal{N}_5/(h\xi^4) \geq 1.28$ for $\xi \leq 0.3$.

The region $\xi \geq 0.3$. Evaluate $\operatorname{Br}_j/\xi^{p_j}$ in interval arithmetic on boxes covering the part of D_j with $\xi \geq 0.3$, starting from a 64×64 grid and bisecting any box whose lower endpoint is not positive. The procedure terminates after 4100, resp. 4124, box evaluations, with 1, resp. 7, boxes of the initial grid bisected once; the final cover consists of 4099, resp. 4117, boxes, all with positive lower endpoints. $\square$

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